Skills, Tasks and Degrees

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August 22, 2025

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Abstract

An increasing number of young university graduates are not entering professional occupations, raising concerns about a decline in their skills compared to earlier cohorts. This perspective may overlook evolving occupational returns to skill and job amenities. I develop an economic model featuring heterogeneous skill supply, differentiated returns to skill across occupations, and changing preferences over non-wage job characteristics. Estimating the model using UK data from 2001–2019, I find a significant decline in average graduate skill levels - about 42% of a standard deviation - which explains the majority of reduced professional employment. Additionally, changes in amenity values push graduates into routine occupations, while lower returns to skill in service occupations create growing demand from low skilled graduates.

Keywords: skills, tasks, tertiary education, occupation choice
JEL Classification: I23, I24, I26, J24

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I would like to thank Konstantinos Angelopoulos, Richard Foltyn, Spyridon Lazarakis, Margaret Leighton, and Jim Malley for comments on earlier versions of this paper. I further thank the seminar attendants at Durham and Lancaster University, the EEA-ESEM Congress (2024), the European Society of Labour Economists Conference (2024), and the 2nd Toulouse Economics of Education Workshop (2025) for their insights. All remaining errors are my own. This research was enabled by the Hamilton HPC Service of Durham University.

1 Introduction

In the UK, increasing numbers of young graduates fail to secure employment in a job where a degree is a required qualification. This underemployment, together with rising graduate earnings inequality, cast doubt on whether universities are delivering on their promises of *graduate skills* and *graduate jobs* for their alumni (see Lindley & McIntosh, 2015, Altonji, Arcidiacono, et al., 2016, Hou, 2023). In the UK and elsewhere, these developments have meant that the value of a university degree is coming under increased public scrutiny, with some questioning the return on this costly investment made by many young individuals.[1]

At the same time, record numbers of high school graduates apply to and are accepted into university. Over the last 30 years, the UK, like other developed economies, has experienced a rapid expansion of tertiary education participation.[2] This has had a strong impact on the composition of the workforce in the UK, particularly among those who are just entering the labour market: Between 2001 and 2019, the share of young adults, aged 21 to 30 with a university degree approximately doubled from around 20% to almost 40% (see Figure A1). Early career mismatch can have long run implications for lifetime earnings trajectories potentially leading to persistent underemployment and skill mismatch (Liu et al., 2016, Erdsiek, 2021, Dickson et al., 2024). This has important implications for the individuals affected and the economy as a whole.

Declining graduate quality is often (anecdotally) cited as a main cause behind rising levels of graduate underemployment. But, the changing supply of skills is only one part of the ever-developing labour market. Over the past decades, there have been significant changes in the demand for the skills of young graduates, driven by changes in technology and the wider macroeconomic environment (see Acemoglu & Autor, 2011 for a survey). Furthermore, current generations likely have different preferences regarding non-pecuniary job characteristics, which evolve over time and influence occupational choices. The observed patterns of labour market outcomes for young graduates are the result of the interactions between these forces.

The question, therefore, is to what degree the observed patterns in the labour market outcomes of young graduates are due to the changing distribution of graduate skills, the evolving patterns of returns to these skills, or changing preferences over non-wage attributes of different occupations? Understanding these different factors is crucial for developing effective policy interventions, as they help identify the key drivers behind

[1] The topic is readily discussed across major news sources and opinion pieces, such as the Financial Times or The Guardian.

[2] Since the passage of the Further and Higher Education Act 1992, university enrolment has roughly doubled to approximately 2.5 million in 2019/20, a trend that was sustained in the face of stark tuition fee increases. The cap on the amount that universities can charge was increased nearly threefold in England in 2012, leading to a large increase in tuition fees, with most institutions charging the maximum amount.

graduate underemployment and wage inequality, allowing for more targeted policy interventions.

A key challenge when it comes to answering this question is that labour market skills are generally unobservable for the econometrician. To address this issue, this paper takes another approach - framing the question as a latent variable problem: skills are unobserved but related to observable choices and labour market outcomes. I develop a model of occupational choice for university graduates and use it to quantify the importance of the demand for and supply of graduates' skills for the labour market outcomes of young graduates.

I structurally estimate the model using a sample of recent UK university graduates from 2001-2019 and recover the parameters of the underlying latent skill distributions for different cohorts of university graduates, as well as the changing returns to skills across different occupations. I then use the estimated model to analyse changes in the graduate skill distribution, the changing returns to these skills, and their combined effects on labour market sorting among university graduates over the first two decades of the 21st century.

I find that between 2001-05 and 2016-19, the mean of the distribution of graduates' skills has decreased by about 42% of a standard deviation, with a particularly strong increase in the share of low-skilled graduates. This had a direct effect on the share of young graduates entering professional occupations, explaining the majority of the decline observed in the data.

At the same time increasing returns to skill in professional occupations leads to stronger sorting pressure, which induces some graduates to instead sort into routine occupations, where the reward for skill has also grown since the early 2000s. Both in terms of returns to skill and non-wage amenities, routine jobs in the late 2010s are comparable to professional occupations in the early 2000s, providing an attractive destination for graduates. The developments in routine and professional occupations differ starkly from those in service occupations, where returns to skill have fallen since 2005.

I decompose the observed trends through the lens of the model to show that the supply of graduates' skills, the return to those skills and non-pecuniary elements of occupations are all important factors in determining the occupation choices of university graduates. Addressing the overall question of graduate underemployment requires considering each of these factors as well as their interactions.

1.1 Related literature

This paper adds to the literature on graduate underemployment with a specific focus on the UK (see Green & Zhu, 2010, Holmes & Mayhew, 2016, O'Leary & Sloane, 2016). These studies find that graduates in the UK are increasingly likely to be employed in roles that were not traditionally considered graduate occupations. I complement their findings by providing a flexible occupational choice framework that can accommodate three potential drivers of these trends: i) changes in the distribution of graduates' skills as a result of higher education expansion; ii) changes in the return to graduates' skills in different occupations due to technological change; iii) changing non-pecuniary preferences over different occupations. My main findings suggest that while changes in the distribution of skills play a major role in explaining a reduction in the share of graduates entering professional occupations, they are not the main driver of the reallocation of these graduates towards routine occupations. Differences in preferences for non-pecuniary aspects of occupations, on the other hand, can explain a substantial amount of the reallocation towards routine occupations. At the same time, declining returns to skill in service occupations makes these a particularly attractive choice for lower skilled graduates.

This paper also contributes to a large literature on the returns to higher education with an emphasis on the heterogeneity of returns (see Altonji, Arcidiacono, et al., 2016, Leighton & Speer, 2020, Andrews et al., 2024 and Lovenheim & Smith, 2022 for extensive surveys). Generally, these studies aim to estimate the returns to post-secondary education while trying to address the inherent difficulties caused by selection effects across dimensions of inherent ability and preferences, using administrative cut-off rules (see for example Hastings et al., 2013, Kirkeboen et al., 2016) or attempting to control for observable factors (Hamermesh & Donald, 2008). A general finding of this literature is that returns to a college degree vary according to several factors, such as the field of study, degree classification earned, or institution attended. In this paper, I focus on the interaction between the supply of and return to graduates' skills in a partial equilibrium framework, thereby providing a potential mechanism for the observed differences in labour market outcomes for different cohorts of graduates.

Furthermore, this paper also speaks to a large body of literature that investigates how technological change affects the sorting of different workers across occupations and, correspondingly, the wage distribution. Exemplified by the task framework based on the seminal work of Autor, Levy, et al., 2003 and further developed in Acemoglu & Autor, 2011, the task framework shifts the emphasis onto specific job tasks and provides an explanatory framework in which a worker's skill set and the tasks to which they are assigned are jointly important for individual productivity and earnings. This provides an incentive to consider changes in the supply of and the demand for skills as important drivers of labour market sorting. While earlier work has focused primarily on the demand for skills (c.f. Firpo et al., 2011, Autor & Handel, 2013, Goos et al., 2014, Deming, 2017, Henseke et al., 2025), recent work has begun to look at the supply and demand of skills jointly (see Roys & Taber, 2022, Diegert, 2024). Considering both skill supply and demand allows for a more comprehensive understanding of labour market dynamics, as it captures not only the availability of skills but also how they are valued across different occupations. Accounting for these forces separately is particularly relevant when considering long time horizons where the skill distribution might have changed over time, and is important for assessing and shaping policy that influences the distribution of skills. My paper represents another attempt in this growing literature, based closely on the approach outlined in Diegert, 2024, who looks at the population of workers in the US, while this paper is focussed on the specific question related to recent university graduates in the UK.

Finally, this paper complements previous work that aims to elicit the skill content of university degrees. Altonji, Kahn, et al., 2014 create measures of the task content of different subjects by mapping task measures from the Dictionary of Occupational Titles to graduates' occupation choices. Similarly, Hemelt et al., 2023 collect information from online job postings to associate desired skills with different degree subjects. My paper differs in that it uses occupation choice and wage information to estimate a continuous distribution of graduate skills. By focusing on wage outcomes and occupational sorting, my approach provides a more dynamic picture of how skills translate into labour market performance, taking into account both monetary rewards and the heterogeneity in how skills are valued across occupations.

To the best of my knowledge, this paper is the first to estimate the returns to and supply of university graduates' skills in the UK and apply the model to the question of graduate underemployment.

The rest of the paper is structured as follows: Section 2 presents some motivating facts about the labour market outcomes of young graduates in the UK over the last 20 years; Section 3 outlines the economic model of wage-setting and occupational choice; Section 4 presents the econometric strategy used to estimate the parameters of interest; Section 5 covers the discussion of results and Section 6 presents the counterfactual decompositions; finally, Section 7 concludes.

2 Motivating empirics

The following section provides some evidence on how labour market outcomes for young graduates have evolved over the last 20 years, highlighting significant shifts that have implications for current debates on the effectiveness of higher education and the value of a university degree in today's economy. My main data source is the Quarterly Labour Force Survey (QLFS), which is a representative household survey in the UK, surveying approximately 40,000 responding UK households per quarter. The survey features a staggered longitudinal design, where households are interviewed for five consecutive quarters and

20% of the sample is replaced in every wave.[3] In this section, I will use the five-quarter longitudinal version of the QLFS to motivate the analysis in the rest of the paper.

In my analysis, I focus on the outcomes of *young* graduates aged between 21 and 30 years.[4] This group is likely more homogenous, and also has less labour market experience, meaning that their skill endowments will be more closely related to their post-graduation endowments, which are of key interest to this paper. Furthermore, the outcomes of this age group are shaping the public debate around university effectiveness.[5]

Figure 1 presents the occupation shares of young graduates between 2001 and 2019 disaggregated by 3 major occupation groups - professional, routine and services, as well as non-employment.[6] By most accounts, a *graduate job* is a key marker of success for a young graduate, commonly identified with employment in professional occupations such as Law, Medicine, or Financial Services. The data reveals that a majority of young graduates are employed in professional occupations, with a high of around 65% in the early 2000s. The share exhibits a declining trend into the post-financial crisis years, bottoming out around 56% in the early 2010s, after which the numbers rebound somewhat. In contrast, the share of young graduates working in routine and service occupations increases steadily over the sampled period. Employment in routine occupations begins at around 15% in the early 2000s, but rises steadily over the sample period, increasing by around 4 percentage points (pp) by the end of the 2010s. The rise in the share of routine workers among graduates is particularly interesting, because over the same time period, the share of routine workers amongst young adults without a degree is falling (see Figure A2 in Appendix A). The share of graduates in service occupations likewise rises by around 4 pp over the 19-year period, although the lower initial baseline means that this represents an increase of close to 50%. Finally, the share of young graduates not in employment peaks around the years of the Great Recession (2010-2012) but then falls quite quickly to below its initial level.

Figure 2 shows trends in hourly real wages of young graduates over the same period. Those employed in professional occupations earn by far the highest wages, as expected, but there is a general decline in hourly wages across all occupation groups in the wake of the financial crisis. The wages of professionals contract most sharply between 2009 and 2015, which is in stark contrast to the wages of those employed in routine occupations, which only experienced a small decline before making a strong recovery. Wages for service occupations fall quite notably in the first decade of the sample but then rebound fairly

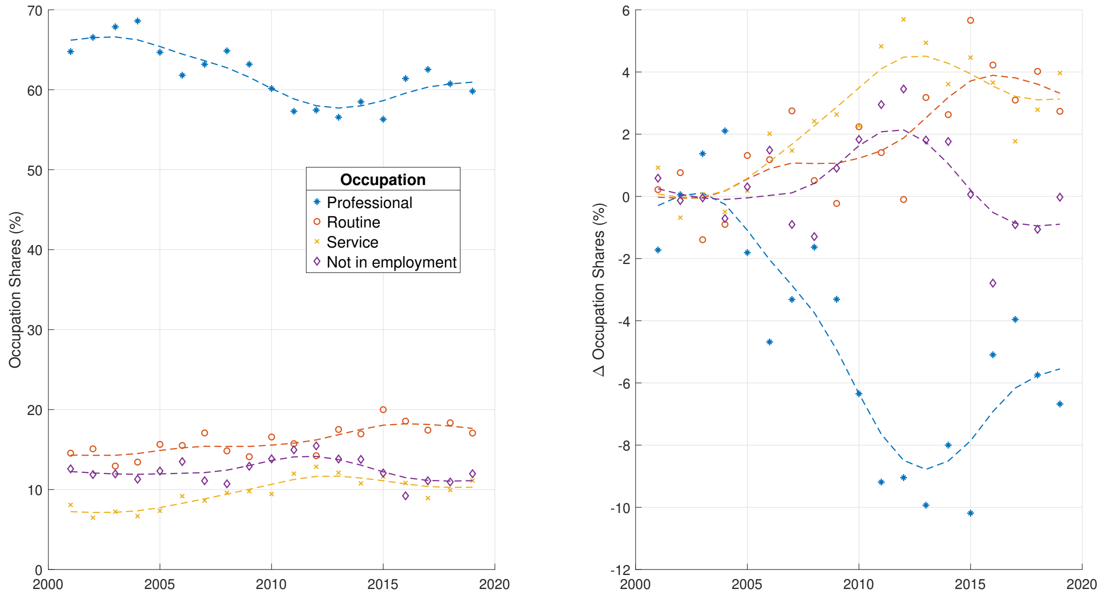
[3]In this respect, the QLFS is similar to the CPS, albeit with a quarterly frequency.

[4]Typically in the UK, students finish high school at 18 and enter 3-year University Courses.

[5]Often, outcomes 5 years post-graduation are seen as an important milestone to judge the success of young graduates.

[6]Occupational groups are based on 1-digit SOC 2000 classifications. Professionals include codes 1-3, routine includes codes 4, 5, 8 & 9, and service includes codes 6 & 7. Non-employed combines the unemployed and those not in the labour force for other reasons. SOC 2010 codes are recoded to SOC 2000 using a crosswalk.

Figure 1: Trends in occupation shares of young graduates in the UK (2001–2019)

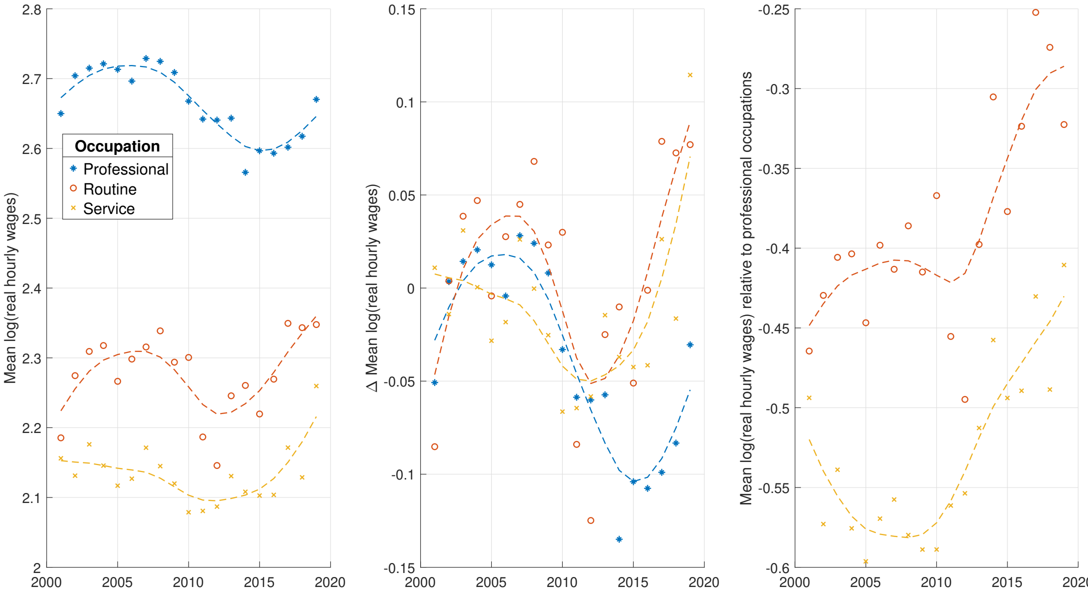


Note: Graduates aged 21-30 years; change relative to 2001-05 average. Broken lines are HP-filtered trends (smoothing parameter = 6.25). SOC 2000 1-Digit Occupation Classification. Professional includes codes 1-3; Routine includes codes 4,5,8,9; Service includes codes 6,7; Not in employment includes unemployed and those out of the labour force for other reasons. *Source:* Quarterly Labour Force Survey (2001-2019).

strongly in the second half. Indeed, while wages in professional occupations have not recovered their early 2000s levels, graduates in routine and service occupations have seen their wages recover much more quickly from the financial crisis and its aftermath. As a result of these dynamics, the relative pay premium enjoyed by those in professional occupations has fallen by about a third relative to routine occupations and about a fifth relative to service occupations.

Apart from changes in relative wages between different occupations, there also have been substantial movements in the distribution of wages within occupations. Figure 3 plots three measures of wage inequality for the three occupation groups, namely the coefficient of variation of log hourly wages, and the P50-P10 and P90-P50 measures of lower and upper tail inequality. Over the time period in question, overall wage inequality has risen slightly for professional occupations, and quite strongly for routine occupations. Service occupations, on the other hand, appear to exhibit a strong U-shaped trend, where wage inequality initially falls until around 2010, after which it begins to rise again. These trends in overall wage inequality are related to changes in the lower and upper tail measures of wage inequality. Lower tail inequality, as measured by the P50-P10 measure, trends downwards for all occupations over the sample period, although there is a distinct hump-shaped pattern for routine occupations. This suggests that inequality at the lower end of the within-occupation wage distribution is broadly falling over the sample period, even though this trend is not very pronounced among professional and routine occupations. Looking at changes in upper tail inequality, there is a slight downward trend among

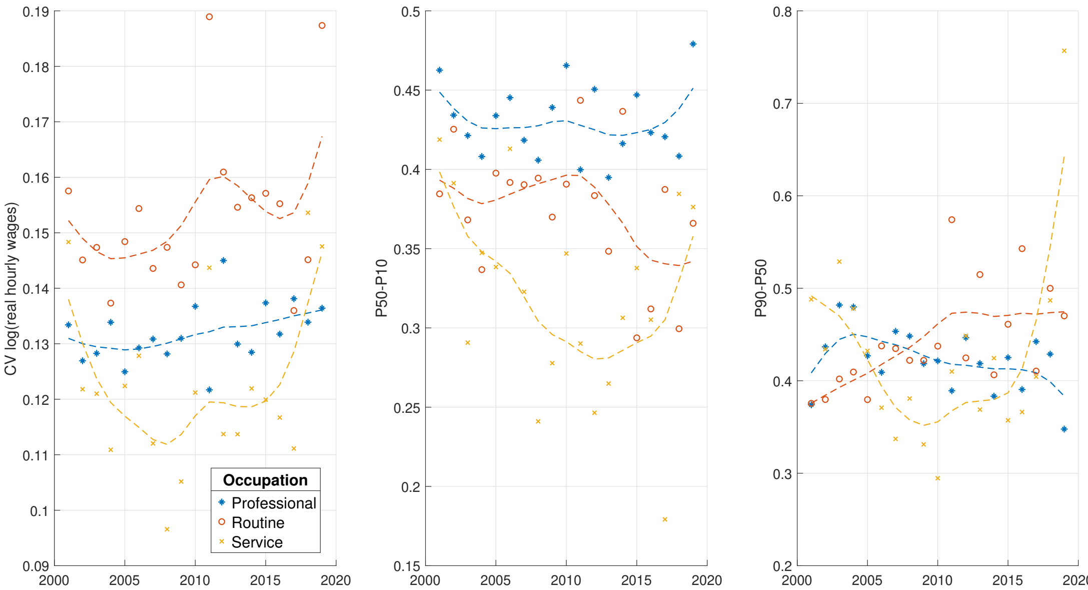
Figure 2: Trends in real wages among young graduates in the UK



Note: Wages are log hourly wages, deflated by 2014 CPI. Working graduates aged 21-30 years. Broken lines represent HP-filtered trends (smoothing parameter = 6.25). Change relative to 2001-05 average. SOC 2000 1-Digit Occupation Classification. Professional includes codes 1-3; Routine includes codes 4,5,8,9; Service includes codes 6,7. *Source:* Quarterly Labour Force Survey (2001-2019).

professionals, but a pronounced upward trend among routine workers. Service workers see a strong U-shaped pattern that somewhat mirrors the picture already documented for the coefficient of variation.

Figure 3: Trends in wage inequality among young graduates in the UK



Note: Wages are log hourly wages, deflated by 2014 CPI. Working graduates aged 21-30 years. Broken lines represent HP-filtered trends (smoothing parameter = 6.25). Change relative to 2001-05 average. SOC 2000 1-Digit Occupation Classification. Professional includes codes 1-3; Routine includes codes 4,5,8,9; Service includes codes 6,7. *Source:* Quarterly Labour Force Survey (2001-2019).

The data presented in this section highlight some interesting trends in the labour market outcomes of young graduates. Over the two decades preceding the COVID-19 pandemic, the share of graduates entering professional occupations has declined while those of routine and service occupations have grown. At the same time, the wages of graduates in professional occupations have fallen post-2008, particularly relative to wages earned in routine and service occupations. While economic logic dictates that changes in relative wages should result in a reallocation of labour from one occupation to another, the underlying drivers of these changes are not well understood. In the following section, I will outline a quantitative economic model that might shed some light on the deeper, structural forces behind these trends.

3 Model

In this section, I present an economic model of occupation choice and wage determination for university graduates. The economic environment in this model closely follows the literature on task-based occupational choice and wage determination (c.f. Autor & Handel, 2013, Roys & Taber, 2016), while the decision framework of the graduate is based on the approach of the multinomial choice literature (see K. E. Train, 2009, Chapter 6).

My model begins at labour market entry, and abstracts from the decision to enter university, and other decisions taken during higher education. Instead, I assume that all unobserved heterogeneity among graduates can be summarised by a latent vector of unobserved skills s that captures all relevant differences in labour market skills among graduates.[7] Upon graduation, graduates draw a stochastic realisation of their skill set from a distribution. Given a set of skills, a graduate then forms expectations about the wage they can earn across all possible occupations, which depends on the occupation-specific return to skill. They then choose an occupation match, taking into account their expected wage and preferences over non-pecuniary job-attributes. This match is then observed by the econometrician. The detailed timeline assumed to hold in the model is specified in Figure 4 below.

For ease of exposition, I first present a cross-sectional version of the model, with suppressed time subscripts. The extension to a more dynamic framework, where parameters

[7]Since no direct or proxy measure of skill exists in the data, I am unable to utilise an auxiliary measurement equation as for example Heckman et al., 2006. This somewhat limits the interpretability of what exactly labour market skills refer to. However, the structure of the model also restricts the set of possible interpretations. Specifically, based on the model formulation, s is an individual-specific factor that is i) fixed in the short run (5-quarters), ii) creates variation in earnings amongst otherwise observationally equivalent individuals, iii) influences an individual's occupation choice through wages. Given that the paper is focused on university graduates, the most likely interpretation of s is the ability of being able to solve complex analytical problems. This interpretation is supported by the findings on the returns to skill in Section 5.

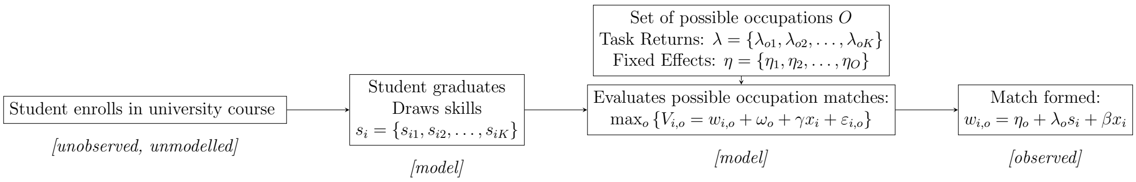


Figure 4: Model Timeline

are allowed to vary over time is straightforward.[8]

A graduate's multidimensional skill set is summarized by a K dimensional vector $s_i = \{s_{i1}, s_{i2}, .., s_{iK}\}$ where each element $s_{ik} \geq 0$ describes how effective graduate i is at performing task k. I assume that s_i is an unobserved, random vector that is drawn from a parametric distribution $s_i \sim S$ that is the same for all individuals $i \in I$. On the demand side, the labour market consists of a large number of *occupations* that each use the different skills supplied to them in different proportions. Specifically, every occupation $o \in O$ has an associated vector $\lambda_o = \{\lambda_{o1}, \lambda_{o2}, ...\lambda_{oK}\}$ where each element $\lambda_{ok} \geq 0$ summarizes the productivity of task k in occupation o.[9]

A graduate's productivity, therefore, depends on her skill-set s_i as well as the task-specific returns λ_o of the occupation she is matched with. Specifically, I assume that the potential log wage of graduate i in occupation o can be written as:

$$w_{i,o} = \eta_o + \lambda_o s_i + \beta x_i, \tag{1}$$

where η_o is an occupation-specific fixed effect, x_i is a vector of other characteristics of the graduate, including sex, potential experience and whether the graduate holds an advanced degree; β is a vector of coefficients.[10] This setup is common in the literature on tasks and skills (c.f. Autor & Handel, 2013, Roys & Taber, 2016).

After graduation, graduates observe their own skills, and all potentially relevant characteristics of all occupations and pick whichever occupation provides them with the highest utility. That is to say that every graduate can observe the set O of all available occupations and attach a personal valuation $V_{i,o}$ to each of these options. I assume that the utility derived from the occupation is linear in the log wage and other occupation characteristics. This is likely to be the case for an economic agent with a suitably defined utility function (e.g. logarithmic), who is borrowing constrained.[11] This leads to the

[8] In the dynamic setup the graduate makes a sequence of occupation-choice decisions, where their level of skill s_i creates correlation across the sequence of decisions.

[9] In the rest of the paper I will refer to these as the *task-* or *skill-specific* returns of occupation o.

[10] In principle the unobserved skill vector s_i should contain all relevant information for the determination of the graduates' wage, but in order to ensure robustness in the empirical application I further control for the sex of the graduate, a second order polynomial of their labour market experience and whether they have an advanced degree.

[11] I believe it reasonable to assume that this situation applies to the sample population studied in this paper.

following potential valuation:

$$V_{i,o} = w_{i,o} + \omega_o + \gamma x_i + \frac{\varepsilon_{i,o}}{\rho}, \tag{2}$$

where o is one of the available occupations, $w_{i,o}$ is the expected log wage earned by i in occupation o, ω_o is an occupation-specific preference term that is common among all graduates, x_i is a vector of other characteristics of the graduate, including sex, potential experience and whether the graduate holds an advanced degree and $\varepsilon_{i,o}$ is an individual-specific preference shock that is *i.i.d.* across all agents and all occupations.[12] $\rho > 0$ is a scalar that governs the relative importance of the idiosyncratic preference shock in the agent's valuation. Accordingly, a worker i solves the following (static) occupational choice problem:

$$V_i = \max_{o \in O} \{V_{i,o}\} \tag{3}$$

Under these circumstances, the individual's occupation choice o_i will refer to the occupation with the highest valuation. Importantly, the value of $V_{i,o}$ is observed by the economic agent, while only o_i^* is observed by the econometrician. For the remainder of this paper, I will denote the observed occupation choice and wage as o_i^* and w_i^* respectively, while their model-implied counterparts are denoted without the asterisk.

Generally, economies of the type described above are characterised by the sorting of workers according to comparative advantage (see Roy, 1951). This self-selection of workers into different occupations according to their different abilities poses the main obstacle that is faced by the literature that is concerned with estimating *task returns* (i.e. the set λ) since there will be a positive correlation between an occupations' task prices λ_o and the skills supplied by workers selecting into this occupation (see Autor, 2013). In this paper, however, rather than being harmful, self-selection is actually helpful as it allows us to make inferences from a worker's observed occupation o_i^* to her unobserved skill-set s_i.

[12] Given that $\varepsilon_{i,o}$ is idiosyncratic, two workers with the same deterministic wage may have different preferences over the set of occupations. This differentiation in choice behaviour is important since otherwise, the utility-maximising choice would be the same for every worker, leading to unrealistic predictions. Furthermore, the introduction of this random term allows us to capture other factors that influence occupation choice besides the desire to maximise wages, such as other preferences or frictions in the labour market.

4 Econometric strategy & estimation

4.1 Econometric strategy

The primary goal of my econometric strategy is to estimate the structural parameters of the model by leveraging both the observed occupation choices and the wages of graduates. The key insight is that a graduate's occupation choice o_i^* and realized wage w_i^* are both informative about their unobserved skill level s_i, given the economic structure described in the previous section. The parameters of interest include those related to the determination of the log wage (η, λ, β), the non-pecuniary utility (γ, ω, ρ), and the distribution of graduates' skills (S).

For estimation, I focus on a model with a single task ($K = 1$), which simplifies the analysis while capturing the essential variation in the data.[13] The labour market is divided into the three major occupation groups, along with the outside option of not working, resulting in four possible occupation choices ($O = 4$). To account for potential changes in the skill distribution over time, I separate the sample of graduates into four cohorts, denoted by $c = 1, \ldots, 4$, based on the year they are first observed in the sample.[14] Each cohort of graduates then corresponds to a cohort-specific skill distribution S_c. Without loss of generality, assume that graduates' skills can take values in the set $s_i \in [0, 1]$. I make no further assumptions about the distribution of S_c, instead, I rely on a novel identification result by Diegert, 2024 to identify S_c non-parametrically.

To account for changes in the returns to skill and job amenities over time, I allow the occupation fixed effects ($\eta_{o,\tau}$), the occupation-specific returns to skill ($\lambda_{o,\tau}$), and the occupation-specific preference terms ($\omega_{o,\tau}$) to vary annually, where $\tau = 2001, \ldots, 2019$.[15] This temporal variation accommodates potential changes in the labour market across different years, which might have influenced graduates' labour market choices and outcomes.

I exploit the panel structure of the Quarterly Labour Force Survey (QLFS) dataset to identify the model parameters. Each individual's choices are observed over $T = 5$ consecutive quarters, providing multiple observations per individual. This panel data structure is crucial, as it allows me to observe within-individual variation over time, which aids in disentangling the effects of unobserved skills from the returns to skill. The

[13]The choice to restrict the model to a single task is driven by a desire to keep the interpretation of skill consistent across cohorts. While in principle the data would allow the identification of two different skills, it would be difficult to give a consistent interpretation of what these correspond to, exacerbating the problem described in Section 3. I also check the assumption of the univariate model using a measurement equation test *ala* Diegert, 2024. Figure A4, shows that the measurements between period t and $t + 1$ are clustered around the 45-degree line for occupation stayers and switchers, suggesting no deviation from the unidimensional model.

[14]Resulting periods cover 2001-05, 2006-10, 2011-15 and 2016-19.

[15]In the following I will refer to time varying variables with the index τ in cases where I refer to different years. I will use t indices, when referring to variables that change from the perspective of the economic agent without reference to a specific time period.

final dataset contains 12,084 graduates, totalling 60,420 observations.[16]

The QLFS records wages only at the first and last interview, so I impute missing wage observations, using a Random Forest regression, controlling for individual fixed effects. This approach preserves cross-sectional wage differences while, by construction, attenuating purely idiosyncratic within-individual noise in the imputed periods. Because the structural estimation treats observed wages as signal plus measurement error (see equation 5), replacing missing values with lower-noise predictions increases the effective signal for identifying occupation effects and returns to skill. The details of the imputation procedure are described in Appendix B.

Given the short time span of the panel, I assume that skills remain constant over the five quarters, i.e., $s_{i,t} = s_i$ for all t. This assumption is reasonable, as significant changes in skill levels are unlikely over such a brief period.

I now reintroduce time subscripts into the notation to reflect the panel data structure. The observed occupation and wage outcomes are vectors of length T: $o_i^* = \{o_{i,1}^*, o_{i,2}^*, ..., o_{i,T}^*\}$, $w_i^* = \{w_{i,1}^*, w_{i,2}^*, ..., w_{i,T}^*\}$. This notation emphasises that I observe a sequence of choices and wages for each individual over time, which is essential for the estimation strategy.

To estimate the structural parameters of the model, I proceed in several steps. First, I derive the conditional probability of a graduate choosing a particular occupation, given their unobserved skill level. Second, I incorporate measurement errors in observed wages. Third, I derive the joint probability of observing both the occupation choice and the wage, conditional on the unobserved skill. Finally, I integrate over the distribution of unobserved skills to obtain the unconditional likelihood function, which depends only on observable variables.

I begin by normalising the utility of the outside option (not working, $o = O$) to zero for convenience. I also make the standard assumption that the idiosyncratic occupation preference shocks ε_{it} are independently and identically distributed (i.i.d.) Type I Extreme Value with variance $\frac{\pi^2}{6\rho^2}$, where $\rho > 0$ is a scale parameter. Under these assumptions, the conditional probability that graduate i chooses occupation o_{it}^* at time t, given their skill level s_i, is given by:

$$\Pr(o_{i,t}^*|s_i) = \frac{e^{\rho\left(\eta_{o^*,t}+\lambda_{o^*,t}s_i+(\beta+\gamma)x_{i,t}+\omega_{o*,t}\right)}}{1+\sum_{o=1}^{O-1} e^{\rho\left(\eta_{o,t}+\lambda_{o,t}s_i+(\beta+\gamma)x_{i,t}+\omega_{o,t}\right)}}, \tag{4}$$

which is the familiar logit probability, having substituted the expected wage 1 into the

[16]The QLFS is not designed to be a panel survey and the 5-quarter longitudinal panel suffers from considerable attrition, resulting in a dataset that is significantly smaller than the purely cross-sectional QLFS. However, the 5-Quarter-QLFS dataset contains longitudinal weights to account for attrition bias, which I include in my estimation.

graduates' decision problem 2.

Next, I account for measurement errors in observed wages. Suppose the observed wage $w^*_{i,t}$ is subject to measurement error:

$$w^*_{i,t} = w_{i,t} + \nu_{i,t}, \tag{5}$$

where $\nu_{i,t}$ is an i.i.d. disturbance term, independent of the worker's occupation choice and distributed normally with mean zero and variance $\phi^2_{o,\tau}$, i.e., $\nu_{i,t} \sim \mathcal{N}(0, \phi^2_{o,\tau})$. This measurement error can be interpreted as an idiosyncratic wage shock that is not anticipated by the graduate at the time of occupational choice.[17]

As a result, we can express the probability of observing the observed wage, conditional on the graduate's skill-set s_i and occupation choice $o^*_{i,t}$:

$$\Pr(w^*_{i,t}|s_i, o^*_{i,t}) = \frac{e^{(-\frac{\nu^2_{i,t}}{2\phi^2_{o,t}})}}{\sqrt{2\pi\phi^2_{o,t}}}. \tag{6}$$

Combining the conditional choice probability with this expression, the joint probability of observing both the occupation choice $o^*_{i,t}$ and the wage $w^*_{i,t}$, conditional on s_i, is:

$$\Pr(o^*_{i,t}, w^*_{i,t}|s_i) = \left(\frac{e^{\rho\left(\eta_{o^*,t}+\lambda_{o^*,t}s_i+(\beta+\gamma)x_{i,t}+\omega_{o*,t}\right)}}{1+\sum_{o=1}^{O-1} e^{\rho\left(\eta_{o,t}+\lambda_{o,t}s_i+(\beta+\gamma)x_{i,t}+\omega_{o,t}\right)}}\right)\left(\frac{e^{(-\frac{\nu^2_{i,t}}{2\phi^2_{o,t}})}}{\sqrt{2\pi\phi^2_{o,t}}}\right). \tag{7}$$

Since I observe multiple periods for each individual, the joint probability of observing the full sequence of choices and wages over T periods, conditional on s_i, is given by the product over time:

$$\Pr(o^*_i, w^*_i|s_i) = \prod_{t=1}^{T}\left(\left(\frac{e^{\rho\left(\eta_{o^*,t}+\lambda_{o^*,t}s_i+(\beta+\gamma)x_{i,t}+\omega_{o*,t}\right)}}{1+\sum_{o=1}^{O-1} e^{\rho\left(\eta_{o,t}+\lambda_{o,t}s_i+(\beta+\gamma)x_{i,t}+\omega_{o,t}\right)}}\right)\left(\frac{e^{(-\frac{\nu^2_{i,t}}{2\phi^2_{o,t}})}}{\sqrt{2\pi\phi^2_{o,t}}}\right)\right). \tag{8}$$

Finally, integrating over the distribution of s leads to the unconditional joint proba-

[17] A potential drawback of the imputation is that estimates of $\phi_{i,\tau}$ are likely downward biased. However, since $\nu_{i,t}$ is realised after the graduate has made their decision, this bias does not affect the main results of the paper.

bility:[18]

$$\Pr(o_i^*, w_i^*) = \int \prod_{t=1}^{T} \left(\left(\frac{e^{\rho\left(\eta_{o^*,t}+\lambda_{o^*,t}s_i+(\beta+\gamma)x_{i,t}+\omega_{o*,t}\right)}}{1+\sum_{o=1}^{O-1} e^{\rho\left(\eta_{o,t}+\lambda_{o,t}s_i+(\beta+\gamma)x_{i,t}+\omega_{o,t}\right)}} \right) \left(\frac{e^{(-\frac{\nu_{i,t}^2}{2\phi_{o,t}^2})}}{\sqrt{2\pi\phi_{o,t}^2}} \right) \right) f(s)d(s) \qquad (9)$$

Standard results (c.f. McFadden & K. Train, 2000) guarantee that we can use the unconditional choice probability in (9) to get consistent estimates for η, λ, β, γ, ω and S using simulated maximum likelihood. In Appendix B, I describe a complete algorithm that can be used to estimate the parameters of interest from this model, using the likelihood function implied by (9).

4.2 Identification

In this subsection, I provide an intuitive explanation of how the key parameters of the model are identified. My identification strategy is underpinned by the formal results presented in Diegert, 2024, who provides a non-parametric identification theorem for a model that nests mine. Like him, I exploit the factor structure inherent in the model, which facilitates the identification of its key parameters (see also Heckman et al., 2006 or Aucejo & James, 2021).

Specifically, the unobserved skill s_i serves as a latent variable influencing both wages and occupational choices. Observed wages can therefore be viewed as measurements of these unobserved skills, with the time-varying shocks $\nu_{i,t}$ acting as i.i.d. measurement errors. To identify the parameters of the log wage equation, I exploit the properties of these measurement errors. Since $\nu_{i,t}$ is assumed to be i.i.d. across time and independent of both the unobserved skills s_i and the occupational choices, the variation in observed wages across different occupations and time periods provides the necessary information to disentangle the effects of the unobserved skills, the occupation-specific returns to skill $\lambda_{o,\tau}$, and the occupation fixed effects $\eta_{o,\tau}$.[19] This allows me to identify the wage equation up to a scalar. I introduce a normalisation by fixing the range of s_i to be between 0 and 1. This normalisation enables the identification of the remaining parameters relative to this benchmark. Because I further observe the same individual in different time periods τ, I can further identify changes in $\lambda_{o,\tau}$ and $\eta_{o,\tau}$ using changes in individual wages over time.

Intuitively, since s_i is fixed for each individual, observing changes in wages, as individuals switch occupations over time, helps identify $\lambda_{o,\tau}$ and $\eta_{o,\tau}$ separately from s_i. This

[18]There is no closed-form solution for this integral, but the integration step can be performed via simulation. In the estimation, I use a set of 100 pseudo-random Halton draws.

[19]In particular, the distribution of wages within a given occupation, conditional on $x_{i,t}$, identifies the quantity $s_i\lambda_o^\tau$, while the difference in average wages across occupations helps identify $\eta_{o,\tau}$.

is similar to estimating occupation-specific fixed effects in a panel-data setting, where within-individual variation aids in separating the effects of permanent and idiosyncratic heterogeneity.

Once the parameters of the log wage equation are identified, differences in occupational choices among individuals help pin down the occupation-specific preference parameters $\omega_{o,\tau}$. Given that the expected wages are determined by the estimated $\lambda_{o,\tau}$ and $\eta_{o,\tau}$, any systematic variation in occupational choices beyond what wages can explain is attributed to the non-pecuniary preferences captured by $\omega_{o,\tau}$.

For a more formal verification, one can show that my model adheres to the assumptions I.1–I.4 inDiegert, 2024, which are sufficient for identification in his framework: Firstly, the specification of the log wage as linear in skill with additive errors satisfies Assumptions I.1 and I.4 in Diegert, 2024. The factors affecting productivity are separated into the time-invariant skill s_i and the time-varying shocks $\nu_{i,t}$, as shown in the log wage equation 1. Moreover, when expressed in exponential form, the productivity function is multiplicatively separable in $\nu_{i,t}$ and s_i, this satisfies the multiplicative separability required by Assumption I.4. Secondly, the assumption that $\nu_{i,t}$ is i.i.d. and independent of s_i and the occupational choices satisfies Assumption I.2, ensuring that the productivity shocks are serially independent and independent of skills and preferences. Thirdly, since the productivity shocks $\nu_{i,t}$ do not enter into the utility function, Assumption I.3 is satisfied. Utilities depend only on expected wages (which are functions of s_i and x_i) and preferences, not on the time-varying productivity shocks. Finally, since $K = 1$ and $T = 5$, I satisfy the condition $K < 2T$, which is necessary for identification.

4.3 Estimation and model fit

For estimation, I approximate the distribution of skills S_c using a histogram.[20] I separate the unit interval into $L = 10$ equally spaced bins, each with an associated probability p_{lc}. I assume that graduates draw a bin with the associated probability and then draw a skill s_i from the range of values within that bin with uniform probability.[21] I maximise the likelihood function using a custom global multi-start algorithm, described in Appendix B.

After estimating the model, I evaluate its ability to capture graduates' occupation choices and wage outcomes. For this purpose, I simulate a representative sample of graduates. Figure 5 below highlights the model fit with respect to the occupation choices of graduates, while Figure 6 shows the model fit with respect to average log wages.

[20]By using a histogram, one can approximate any function arbitrarily closely, by increasing the number of bins L.

[21]This approach to approximating the skill distribution is based on the treatment of the distributions of latent taste parameters in K. Train, 2016.

The model fit for the occupational choices of young university graduates is generally close to the actual data, effectively capturing long-run trends.[22] Specifically, the model performs well in aligning with long-term patterns in employment across different occupations, such as the fall of the share of professional occupations and the steady rise of routine and service occupations.

Figure 5: Model fit - occupation shares

Note: Broken lines represent HP-filtered trends with a smoothing parameter of 6.25, based on a simulated sample of graduates.
Source: Baseline Model.

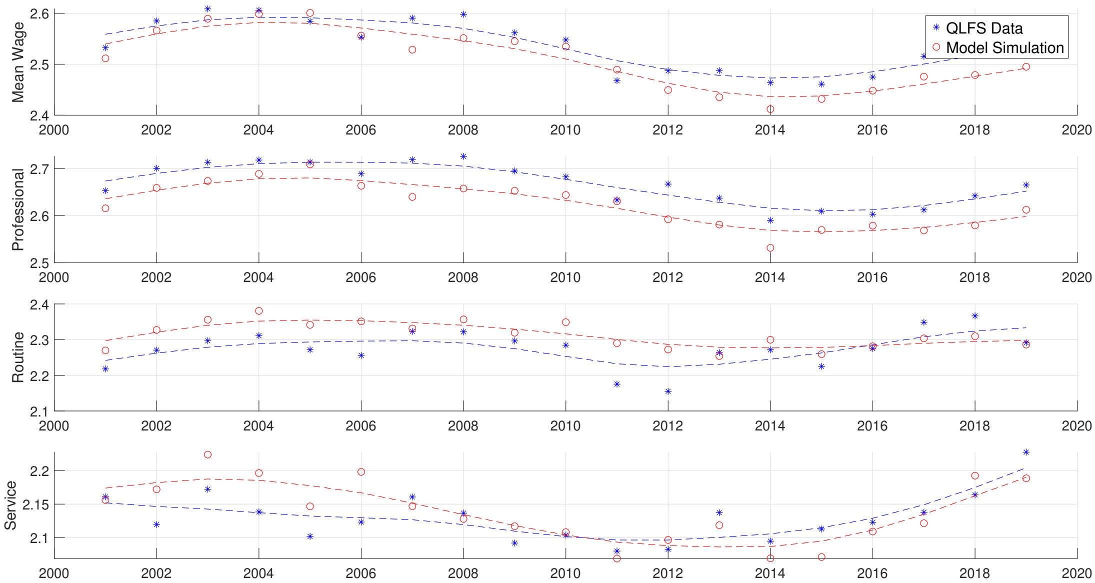
Figure 6 provides a comparison of mean log hourly wages across the occupational categories. The model's simulations align closely with the observed data, reproducing the relative wage differences between occupations and the overarching trends. The model shows a slight under-prediction in the professional category, with an average difference of 0.04 log points in log wages on average. In contrast, the model slightly over-predicts wages in routine and service occupations by 0.04 and 0.01 log points, respectively.

Table 1 shows the detailed model fit across the 4 time periods. It can be seen, that the model tracks the main developments in the data well, capturing for example the decline of the professional share during 2006-15, as well as the slight rebound during the last years of the sample.

Overall, the model does a good job of reproducing the main elements for both occupational distribution and wage outcomes. It effectively captures the observed shifts in employment shares across professional, routine, and service categories, as well as the relative wage dynamics between these groups. In the next section, I will outline the estimation results in some additional detail.

[22]For details see Table A1 in Appendix A.

Figure 6: Model fit - mean log hourly wages



Note: Mean of log hourly wages, deflated by 2014 CPI. Broken lines represent HP-filtered trends with a smoothing parameter of 6.25, based on a simulated sample of graduates. *Source:* Baseline Model.

Table 1: Model Fit by Period

	2001–05		**2006–10**		**2011–15**		**2016–19**	
	Data	**Model**	**Data**	**Model**	**Data**	**Model**	**Data**	**Model**
Share (%)								
Professional	66.260	66.557	62.593	63.088	56.931	55.213	61.544	61.361
Routine	14.705	14.942	16.055	16.628	18.537	18.75	17.682	17.706
Service	7.393	7.447	9.284	8.624	11.392	13.025	10.113	10.589
Non-employed	11.643	11.055	12.068	11.66	13.14	13.012	10.66	10.344
Log Hourly Wage								
Mean	2.587	2.584	2.562	2.537	2.474	2.436	2.514	2.475
Variance	0.151	0.145	0.155	0.153	0.155	0.151	0.141	0.151
P10	2.077	2.104	2.034	2.042	1.954	1.941	2.035	1.991
P50	2.607	2.592	2.584	2.553	2.496	2.455	2.525	2.495
P90	3.066	3.064	3.056	3.015	2.976	2.909	2.975	2.928
Mean Professional	2.706	2.681	2.696	2.647	2.626	2.573	2.624	2.581
Mean Routine	2.277	2.35	2.294	2.335	2.231	2.274	2.335	2.3
Mean Service	2.134	2.187	2.119	2.123	2.105	2.089	2.158	2.153

Note: Simulations based on a representative sample of graduates. *Source:* QLFS (2001-19) and Baseline Model.

5 Results

This section presents the results of the estimated model. I first present the model estimates for the graduate skill distributions, the returns to skill and the non-pecuniary benefits accruing to different occupations, as these are the underlying drivers of the observed dynamics in graduates' labour market outcomes. I then analyse, what the changes in these structural determinants mean for the sorting of graduates into different occupations. Finally, I decompose the importance of the different factors, using counterfactual decompositions in the following section.

5.1 Graduate skills

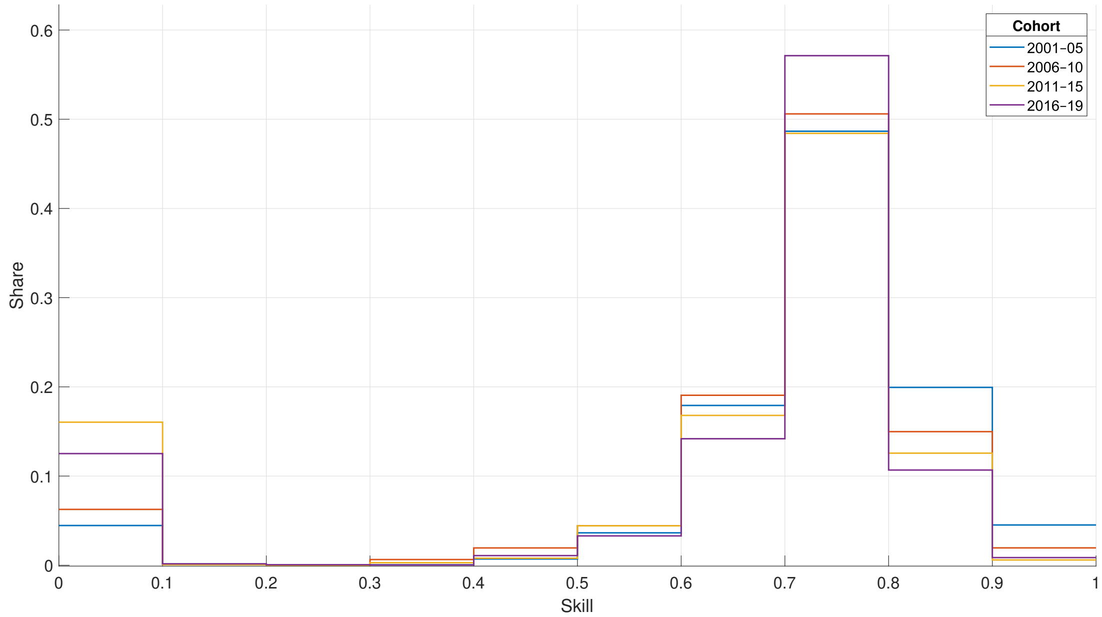
Figure 7 shows the estimated skill distributions for graduates of the four different cohorts.[23] Across cohorts, the distribution of skills shows two notable features: i) The skill distributions appear to be approximately left skewed, with the majority of the probability mass between 0.5 and 0.8. ii) There is an excess mass at the very bottom of the skill distribution. The concentration of graduates among the middle and upper-middle deciles of the distribution of skills, suggests a reasonably homogenous concentration of skill amongst university graduates. Further, the elevated share of graduates in the first decile is indicative of a small number of individuals who somehow fail to acquire skills on par with the majority of their contemporaries. A comparison of the different cohorts suggests that between 2001-05 and 2016-19, the distribution of skills experienced a shift downwards, with a strong fall in the share of graduates in the top 2 skill deciles. Conversely the share of graduates in the middle third of the skill distribution has increased, as well as a considerable increase in the share of graduates in the first decile of the distribution.

Panel A of table 2 allows a more detailed comparison of the different cohorts. The main observation regarding the changing graduate skill distribution across the four cohorts is that between 2001-05 and 2016-19, the mean (median) of the skill distribution fell by 10.1% (2.1%), equivalent to 42% (9%) of a standard deviation among the first cohort, a sizeable reduction. Anecdotally, this shift is consistent with the main reforms to the production of university graduates that occurred over the late 1990s and 2000s. The rapid expansion of higher education participation at both the intensive (increased number of students) and extensive (increased number of HE providers) margins likely negatively affected the average ability of high school leavers entering university, as well as the quality of the instruction they received while at university (see Carneiro & Lee, 2011, Lindley & McIntosh, 2015).[24]

[23]See also Panel B of Table 2.

[24]There is also a perception that students are increasingly avoiding *hard, quantitative* subjects in favour of less *rigorous* degrees, potentially hurting their accumulation of valuable skills. While the longitudinal

Figure 7: Graduate Skill Distributions



Note: Histogram density estimates of cohort-specific skill distributions. *Source:* Baseline Model.

Concurrent with the reduction of average skill level, the degree of skill inequality between graduates increased between the two cohorts. Considering measures of overall variation, the Coefficient of Variation and the Gini index increased by 56.4% and 57.5% respectively. However, these changes are most notable at the bottom of the skill distribution. While the mean-to-median ratio indicates a slight reduction of inequality, the P50-P10 ratio suggests a large increase in inequality at the lower tail of the distribution, which is not mirrored by changes in the P90-P50 ratio, which remains quite stable over the sample period. Other percentile measures support this, as the changes are heavily skewed towards the lower-end of the skill distribution. While the 90th percentile of the 2016-19 cohort is only 6.6% below the 2001-05 cohort, the 10th percentile has fallen by almost 90%. These distributional observations lend further credence to the theory that the main changes to the skill distribution are due to the addition of additional graduates at the left tail of the distribution.

Put into context, these results can already provide a partial explanation as to why the share of professional occupations has fallen among university graduates. Assuming that professional occupations provide a higher return to skills than service or routine occupations, a fall in the average level of skills would - ceteris paribus - make these

version of the QLFS does not include information on the subject of the degree, this information is available in the cross-sectional data. In Figure A3, I plot the time series of the shares of 7 broad subject categories for young graduates. A notable pattern is the steady downward trend among STEM, Business & Economics and Law degrees between 2001 and 2019. If these subjects typically endow graduates with more marketable skills, this would provide corroborating evidence for the possible causes of the changes in the skill distribution.

Table 2: Details of Graduate Skill Distribution

Panel A: Moments of the graduate skill distribution

Levels	$Mean$	$Median$	$\frac{Mean}{Median}$	$Std.Dev.$	$Coef.Var.$	$Gini$	$P10$	$P25$	$P50$	$P75$	$P90$	$\frac{P50}{P10}$	$\frac{P90}{P50}$
2001–05	0.72	0.748	0.963	0.173	0.241	0.109	0.606	0.691	0.748	0.799	0.873	1.235	1.167
2006–10	0.687	0.734	0.936	0.193	0.281	0.128	0.521	0.661	0.734	0.784	0.846	1.409	1.153
2011–15	0.623	0.724	0.86	0.263	0.422	0.202	0.062	0.621	0.724	0.776	0.825	11.612	1.14
2016–19	0.647	0.732	0.884	0.243	0.376	0.171	0.078	0.653	0.732	0.776	0.815	9.342	1.114
Differences	$Mean$	$Median$	$\frac{Mean}{Median}$	$Std.Dev.$	$Coef.Var.$	$Gini$	$P10$	$P25$	$P50$	$P75$	$P90$	$\frac{P50}{P10}$	$\frac{P90}{P50}$
Δ 2006–10	-0.033	-0.014	-0.027	0.02	0.04	0.019	-0.085	-0.031	-0.014	-0.015	-0.027	0.175	-0.015
(%)	-4.539	-1.812	-2.777	11.445	16.744	17.648	-13.971	-4.432	-1.812	-1.843	-3.052	14.133	-1.263
Δ 2011–15	-0.097	-0.024	-0.103	0.09	0.182	0.093	-0.543	-0.07	-0.024	-0.022	-0.048	10.377	-0.028
(%)	-13.502	-3.179	-10.662	51.75	75.439	85.669	-89.705	-10.194	-3.179	-2.811	-5.481	840.454	-2.378
Δ 2016–19	-0.073	-0.016	-0.079	0.07	0.136	0.063	-0.527	-0.038	-0.016	-0.022	-0.058	8.107	-0.053
(%)	-10.14	-2.13	-8.184	40.509	56.364	57.543	-87.064	-5.466	-2.13	-2.813	-6.595	656.595	-4.562

Panel B: Histogram of the graduate skill distribution

Shares	[0–0.1]	[0.1–0.2]	[0.2–0.3]	[0.3–0.4]	[0.4–0.5]	[0.5–0.6]	[0.6–0.7]	[0.7–0.8]	[0.8–0.9]	[0.9–1]
2001–05	0.045	0.001	0	0.001	0.007	0.036	0.179	0.487	0.199	0.045
2006–10	0.063	0.001	0	0.006	0.019	0.044	0.191	0.506	0.15	0.019
2011–15	0.16	0.001	0	0.003	0.008	0.044	0.168	0.484	0.126	0.006
2016–19	0.125	0.002	0.001	0	0.011	0.033	0.142	0.571	0.107	0.009
Differences	[0–0.1]	[0.1–0.2]	[0.2–0.3]	[0.3–0.4]	[0.4–0.5]	[0.5–0.6]	[0.6–0.7]	[0.7–0.8]	[0.8–0.9]	[0.9–1]
Δ 2006–10	0.018	0	0	0.006	0.012	0.008	0.011	0.019	-0.05	-0.026
(%)	40.557	-3.07	63.038	1134.533	177.396	21.895	6.371	3.992	-24.875	-56.862
Δ 2011–15	0.116	0	0	0.002	0.001	0.008	-0.011	-0.002	-0.074	-0.039
(%)	259.959	-36.495	-62.715	451.563	12.768	21.76	-6.229	-0.491	-36.965	-86.821
Δ 2016–19	0.081	0.001	0	0	0.004	-0.003	-0.037	0.085	-0.093	-0.037
(%)	180.942	44.868	230.094	-35.399	54.828	-9.441	-20.833	17.412	-46.466	-80.935

Note: Simulations based on a representative sample of graduates. Changes relative to 2001-05 cohort. Percentage changes in brackets. *Source:* Baseline Model.

occupations less attractive to graduates, leading to a decline in the share of graduates entering these roles.

5.2 Returns to skill and occupational amenities

Figure 8, Subplot 1, provides the model's estimates for the returns to skill for the different occupations. In line with expectations, professional occupations provide the highest return to skills, followed by routine and then service occupations. Overall, the return to skill in routine occupations is much closer to professional returns, while for service occupations, the model estimates close to zero returns from the early 2010s onwards. Negligible returns among service occupations suggest that skilled graduates do not have a particular advantage in these occupations, which might be in line with our expectations given the type of work in these occupational fields.[25] This hierarchy of returns implies that as the overall skill level of graduates decreases, the incentive to pursue professional occupations diminishes, especially for those whose skills no longer meet the higher threshold required. Instead, these graduates may opt for routine or service occupations, where the returns to skill, while lower, are more attainable given their abilities.

Over the 19-year period, there is a pronounced increase in the return to skills across professional and routine occupations, consistent with the theory of within-occupation task upgrading.[26] The trend appears to be stronger in the lead-up to the 2008 financial crisis and then flattens out afterwards. The growth of the return to skill between 2001 and 2010 is 23.3% for professional and 39.8% for routine occupations. In the years between 2010 and 2019, the return to skills in professional occupations increased by 32.1%, but this is mainly driven by a very high estimate for 2019. Excluding this likely outlier the increase is effectively 0 at -0.1%. For routine occupations the trend is still sizeable with an increase of 14.8% between 2010 and 2019.

For service occupations, the trend moves markedly in the opposite direction. While the returns to skill in service jobs, is only marginally lower than that in routine occupations during 2001-05, it quickly diverges after 2006, when returns begin to rapidly fall, hitting 0 around 2012.[27] Returns recover marginally after 2016, but remain very low into 2019.

The differential evolution of the returns to skill provides an additional perspective to the observed trends in graduate's occupation choices. Increasing returns to skill in pro-

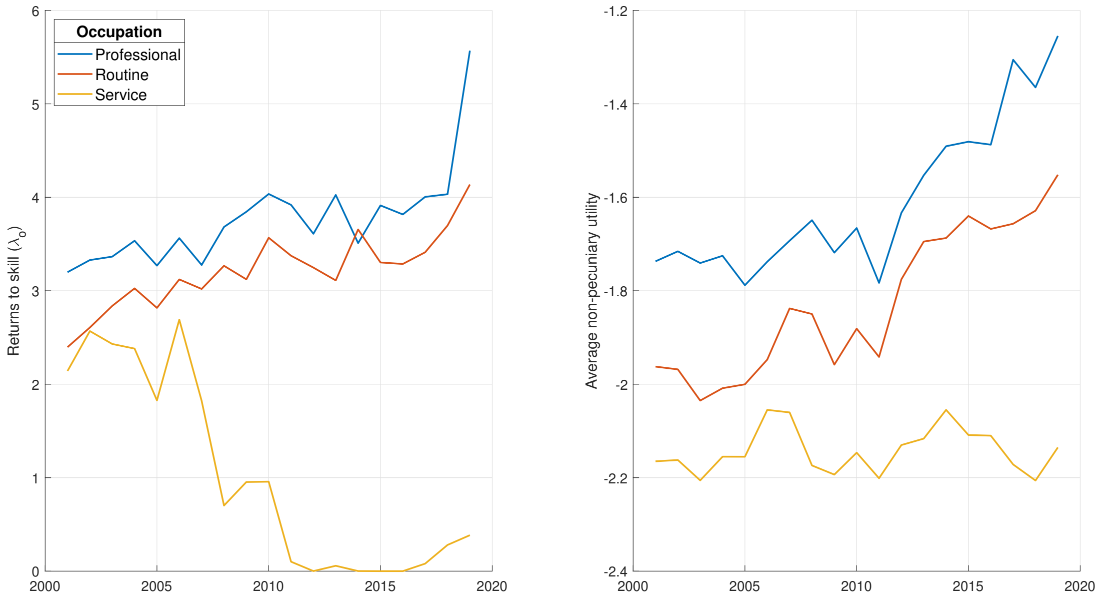
[25]A flat skill-wage profile might also be a result of the wage-setting practices in the majority of service sector jobs, where wages are generally quite rigid, while performance is rewarded through other channels that are unmodelled in this exercise.

[26]Henseke et al., 2025 find some evidence that graduate skill requirements have grown between 1997 and 2017.

[27]While a detailed analysis of the decrease in returns is beyond the scope of this paper, there are a number of possible explanations, such as more binding minimum/living wage in service oriented sectors; automation of customer-facing infrastructure in retail, hospitality and banking or the increasing usage of zero-hours contracts from 2013 onwards.

fessional occupations mean that these occupations become more attractive to graduates, but crucially only those graduates who already exceed a certain threshold level of skill. For less skilled graduates, the higher focus on variable (skill-based) compensation means that professional occupations become less attractive compared to occupations with less variable and more fixed compensation structures. This means that rising returns to skill can actually decrease the share of graduates opting into professional occupations. At the same time, the relative increase in the return to skill in routine occupations has the potential to make these occupations attractive to more skilled graduates. As such routine jobs can become attractive alternatives for graduates at the upper-middle part of the skill distribution. Falling returns to skill in service occupations provide the opposite effect: as returns decrease service occupations become less attractive to medium skilled graduates and more attractive to those with lower levels of skill.

Figure 8: Returns to skill and non-pecuniary utility components by occupation



Note: Subplot 1: Estimates of returns to skill in different occupations. Subplot 2: Estimates of non-pecuniary utility in different occupations averaged over observable covariates. Value of non-employment normalised to 0. *Source:* Baseline Model Estimates.

When choosing an occupation, money is rarely the only objective that matters. Occupations provide important (dis-)amenities to their workers, such as attractive workplace conditions, flexible workdays or a general feeling of prestige associated with a particular job. These additional perks enter the decision framework of workers and can play an important factor in explaining occupation choices in conjunction with differences in earnings and wages (see Sorkin, 2018). The model captures these non-pecuniary preferences in two distinct ways: Firstly, the idiosyncratic preference term $\varepsilon_{i,o}$ that reflects the (random) preferences of individual graduates, and secondly the set of occupation-specific general preferences γ and $\omega_{o,\tau}$ that reflect the general prevailing tastes of the population of graduates as a whole and conditional on their observable characteristics x_i.

Figure 8, Subplot 2, plots the estimated values of the occupation-specific preferences ω_{ot} over the sample period. [28] The plotted values are relative to the outside option of not working, which has been normalised to 0 for all periods. The estimated values are all negative since any work carries a degree of disutility relative to the option of not joining the labour force. Since the numéraire of the model is the log wage, we can deduce that, on average, graduates require a log wage of between 1.2 and 2.2 for them to consider working more attractive than remaining outside the labour market.

While it is generally true that working comes with a utility penalty, there are important variations across occupations and over time. First, there appears to be an ordering among the different occupations, with professional occupations providing the lowest level of disamenity, followed by routine occupations, which dominate service occupations. This pattern is consistent with expectations: professional jobs provide more interesting work, greater autonomy, and are less likely to involve unpleasant or dangerous activities. Second, while the relative rankings remain stable over time, there is a general trend of increasing amenity values across professional and routine occupations from about 2010 until the end of the sample period. During this general rise, the gap between professional and routine occupations appears to narrow somewhat, suggesting that the latter are becoming more attractive relative to professional occupations. Over the same time period, the average non-pecuniary utility derived from working in service occupations, remains approximately stable, which opens a sharp divide relative to professional and routine occupations.

For professional and routine occupations general increase in amenity values could suggest an overall improvement in working conditions, or, since the values are normalised relative to not working, a depreciation of the outside option. Possible drivers of this depreciation could include economic insecurity, such as a lack of stable income for those not in employment, reductions in welfare benefits, or an increase in the perceived risks associated with being out of the labour force for an extended period. Such factors make the option of not working less attractive, thus increasing the relative appeal of available good job opportunities. These explanations are broadly in line with the changes to the welfare system in the UK under the coalition government, such as the introduction of universal credit in 2010. Further, the experience of the financial crisis and the Great Recession would have made job security and stability more relevant to the decision-making of young graduates.

A reduction in the relative gap between professional and routine occupations hints at a relative appreciation of the utility associated with these occupations, possibly through improved working conditions such as better health and safety standards or more flexible

[28] Values are averaged over observable covariates. For details on the effect of demographic characteristics see Figure A6 in Appendix A.

working hours. It could also indicate a more general shift in graduates' attitudes towards such non-graduate jobs, where these roles are increasingly seen as viable and respectable career options. In either case, this reduction in the disamenities gap is likely a contributing factor to the increasing share of graduates in routine occupations, as the improved perception and conditions of these roles make them more attractive to graduates who may otherwise have pursued traditional professional careers.

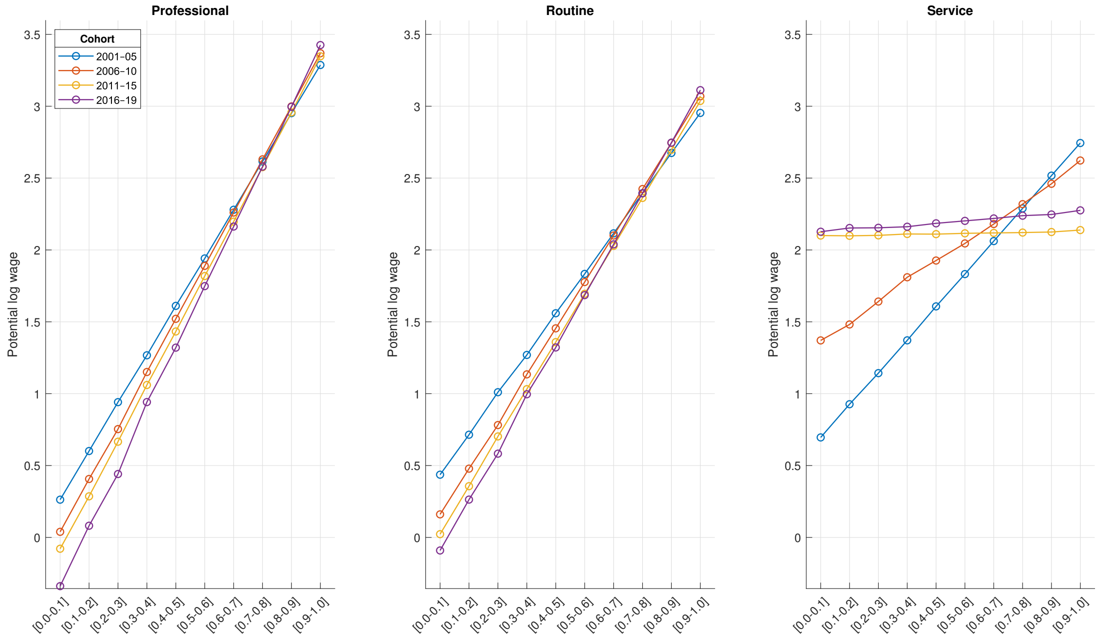
5.3 Skills and sorting

The economy described by the model in Section 3 is characterised by the endogenous sorting of graduates into different occupations. Graduates with high levels of skill will - ceteris paribus - enter occupations where the return to these skills is high, while those with low levels of skill will be drawn to occupations where the skill-based compensation is lower. Figure 9 shows the mean potential hourly wage by skill deciles for the 3 different occupation groups by cohort. Wages increase monotonically by skill deciles for all occupations; however, the rate of increase depends on the value of λ_o and hence the increase is steepest for professional occupations, followed by routine occupations. Service occupations have a much flatter skill-wage profile, particularly for latter cohorts when returns are close to 0.

Generally, the steepness of the log hourly wage curves is an indication of how much sorting pressure is applied across the skill distribution. Very steep profiles indicate a large potential wage difference between high skilled and low skilled graduates, leading to more sorting along the skill distribution. On the other hand, flatter profiles lead to less sorting and self-selection amongst graduates. In light of this, we note that across cohorts, the wage profiles of professional and routine occupations become steeper, leading to more selection, while service occupations become less selective with a wage-skill gradient that becomes almost horizontal in the latter parts of the sample.

Figure 10 illustrates how the potential wage schedules together with the non-pecuniary amenities shape the sorting behaviour of graduates into different occupations. At the lowest skill deciles, graduates do not have a strong incentive to enter professional occupations, or indeed any form of employment. If they consider employment at all, they on balance favour service occupations where the flat skill-wage profile offers at least some monetary compensation. This dynamic results in upward sloping occupational choice curves for professional and routine occupations and falling ones for non-employment, with service occupations presenting a somewhat hump-shaped pattern for the first 2 cohorts and a downward sloping one for the latter cohorts. These patterns are indicative of strong sorting pressures across the skill distribution. For example, a graduate in the top decile of the skill distribution is about 80 pp more likely to enter a professional occupation than one in

Figure 9: Potential log hourly wage by skill deciles

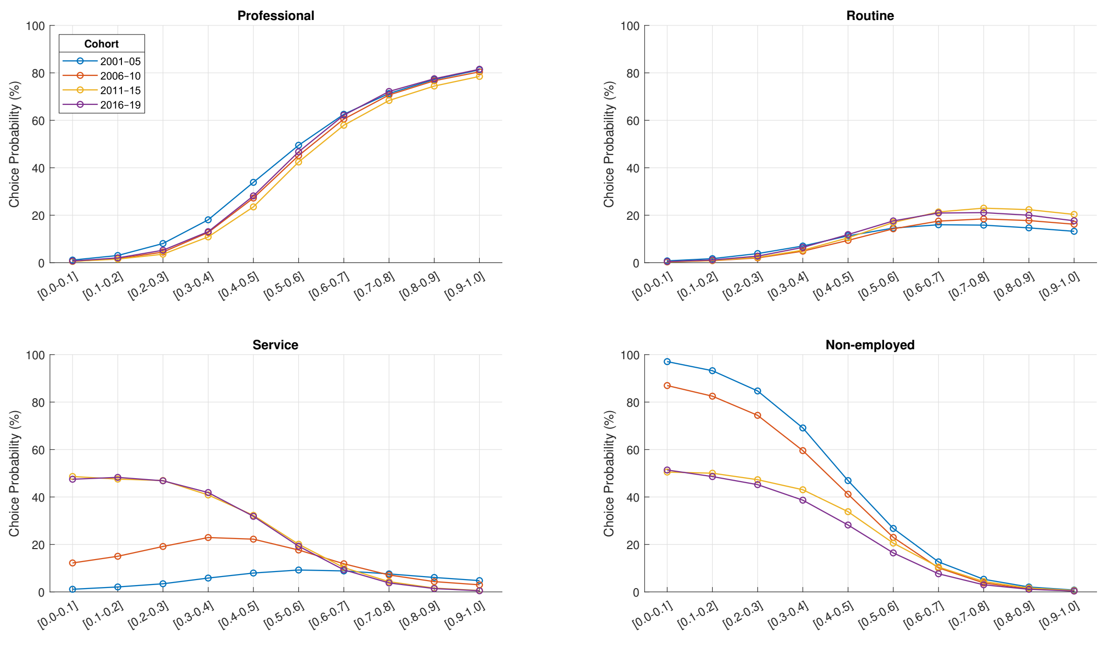


Note: Based on a simulated sample of graduates. *Source:* Baseline Model.

the lowest decile. Similar pressures exist for routine occupations, where the difference is smaller, but still sizeable at 20 pp. With the exception of the first cohort, service occupations tend to attract graduates from the lower part of the skill distribution. Particularly in the latter cohorts, where the potential wage gradient is flat, lower skilled graduates are much more likely to enter service occupations than high skilled ones, with a difference of around 50 pp between the lowest and the highest decile. Likewise graduates choosing to remain out of the labour market are heavily self-selected from the lower part of the skill distribution. For highly skilled graduates the opportunity cost of not working is high, leading most of them to enter the labour market. With lower levels of skill the trade-off become less severe, leading to higher rates of non-employment, with the differential between the top and bottom skill decile reaching about 90 pp among the earlier cohorts.

Figure 11 provides a more detailed assessment of the differences in sorting patterns across the different cohorts. The first panel shows a broad-based decline in conditional choice probabilities for cohorts after 2001–05, with the largest drops concentrated in the middle deciles (roughly 3rd–7th). Movements at the very bottom and top deciles are small by comparison. This pattern implies that access to professional roles has shifted upward in the skill distribution: mid-skill graduates who would previously have cleared the entry threshold are increasingly displaced into other occupations, while top-decile graduates are largely insulated. In model terms, this is consistent with a steeper return-to-skill schedule in professional jobs—raising the payoff to being in the top tail and tightening selection around the margin—so the reallocation pressure is strongest where the marginal graduate

Figure 10: Occupational sorting by skill deciles



Note: Based on a simulated sample of graduates. *Source:* Baseline Model.

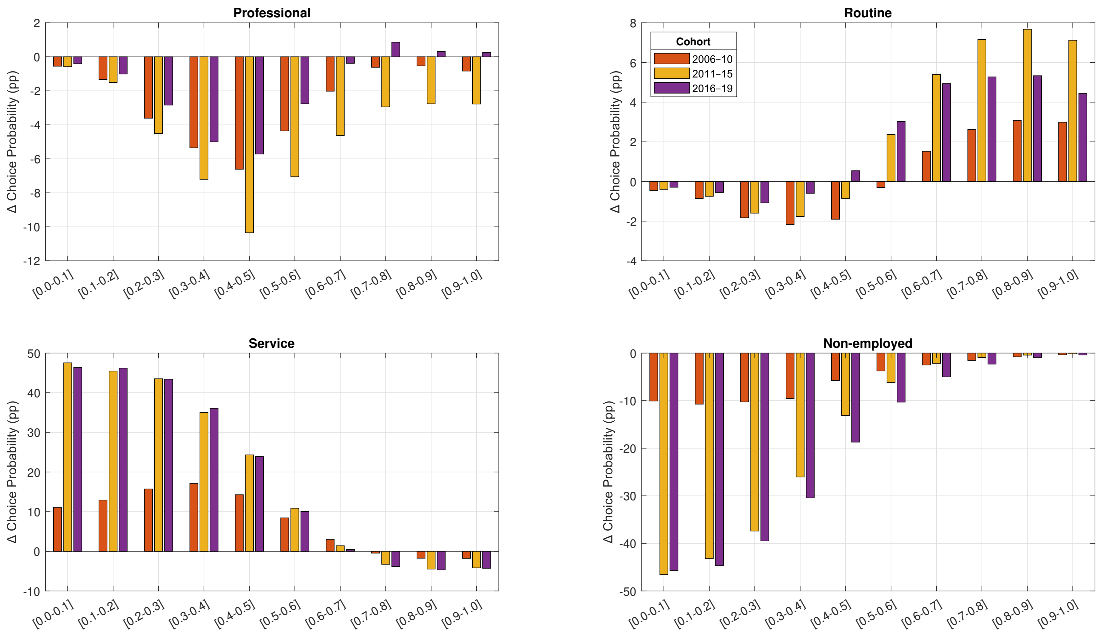
is located.

The second panel reveals a clear pivot in choice probabilities: from the 6th decile onwards, conditional choice probabilities rise sharply for the cohorts after 2001-05. Below this cut-off they edge down, with the decline flattening toward the very bottom. This split pattern indicates stronger sorting by skill: as returns to skill steepen across routine jobs, these become relatively more attractive to higher-skill graduates, boosting choice probabilities in the upper deciles. Symmetrically, lower-skill graduates face weaker incentives to select into routine work because the skill threshold within routine roles has crept up.

The third panel, depicting service occupations, mirrors the trends observed in the previous panel. Relative to 2001–05, conditional choice probabilities jump sharply in the lower half of the skill distribution, especially in the first three deciles, and remain positive but taper through the middle deciles, and then turn mildly negative from about the seventh decile upward. The pattern is most pronounced for the 2011–15 and 2016–19 cohorts but still evident as early as 2006-10. The pattern reflects a flattening of the skill–wage profile in service jobs, which raises their relative attractiveness to lower-skilled graduates while making them less appealing to higher-skilled graduates.

The last panel shows clear declines in conditional choice probabilities for non-employment, concentrated almost entirely below the sixth decile, with the steepest drops occurring from the 2011–15 cohort onward. This indicates that, conditional on skill, graduates in more recent cohorts are much more likely to be employed than their earlier counterparts. The pronounced reduction in the lower deciles aligns closely with the concurrent rise in choice probabilities for service occupations, suggesting that many lower-skilled graduates who might previously have remained out of the labour force are instead entering service jobs.

Figure 11: Changes in occupational sorting by skill deciles



Note: Changes in occupation shares relative to 2001-05 cohort, based on a simulated sample of graduates. *Source:* Baseline Model.

In this section I have documented a number of key findings from the estimated model, namely the evolution of the graduate skill distribution, the returns to skill and non-pecuniary amenities, and how these interact to shape the sorting of graduates into different occupations.

First, the distribution of graduate skills has shifted downward between the 2001–05 and 2016–19 cohorts, with the mean falling by the equivalent of 42% of a standard deviation and the median by 9% of a standard deviation. This change is driven primarily by an increase in the mass of graduates in the lower tail of the distribution, alongside a reduction in the share of graduates in the top two skill deciles. The result is a more compressed upper tail and a more heavily populated lower tail.

Second, returns to skill have diverged sharply across occupations. Returns have risen in professional and routine jobs, while falling substantially in service occupations. These changes translate into steeper potential wage–skill profiles for professional and routine jobs, increasing the sorting pressure by skill, and an almost flat wage–skill profile in service jobs, which reduces their attractiveness to high-skilled graduates.

Finally, conditional on skill, there is evidence of a reallocation of graduates across occupations. Medium- and high-skilled graduates have increasingly entered routine occupations at the expense of professional jobs, consistent with the tighter selection into professional roles. At the lower end of the skill distribution, the main shift has been away from non-employment and into service occupations, reflecting both the reduced relative appeal of not working and the relative accessibility of service jobs for less-skilled graduates.

In the next section, I will assess the relative importance of the different factors quantitatively, by employing some counterfactual decompositions.

6 Counterfactual decompositions

In this section, I consider a number of counterfactual experiments in order to assess the importance of different structural forces in driving the changing labour market outcomes of young graduates. The model allows me to decompose the observed changes by fixing certain parameters at their earlier values and simulating the model into later periods. The differences between these counterfactual simulations with respect to the estimated full model will provide some insight into the underlying factors driving the observed patterns in the data.

There are four main structural forces that I have modelled and that I consider in this exercise: i) the skill distribution of young graduates S_c; ii) the occupation-specific skill prices and returns to skill (represented by η_o and λ_o); and iii) the non-pecuniary component of utility (ω_o and ρ_c) and iv) the changing demographic composition of graduation cohorts (x_i).[29] Each of these four represents a potential mechanism for the changing labour market outcomes of young graduates.

My main aim is to assess how much each of these channels has contributed to the changing occupational distribution of young graduates since the early 2000s, which I why I choose the period 2001-05 as a reference period. To quantify the contribution of each mechanism I employ a Shapley decomposition method (see Shorrocks, 1982).[30] Intuitively the decomposition proceeds, by iteratively fixing a set of parameters at their 2001-05 values and then simulating the model forward under this new set of parameters, comparing the differences in outcomes relative to the baseline model. The decomposition then calculates the marginal contribution of each factor by considering all possible combinations, ensuring a fair allocation of the total effect among the factors. This is particularly rele-

[29]The parameter vectors β and γ are time invariant, so I do not consider them in the decomposition analysis directly. their effect will make itself noticeable through the changing demographic composition of the cohorts.

[30]The model is highly non-linear, with strong interaction effects between the different component parts, which puts more traditional decomposition methods such as those proposed by Blinder, 1973 and Oaxaca, 1973, or more recently Gelbach, 2016 on the back foot. I therefore opt for the Shapley decomposition as a robust option in this setting. Figure A7 in Appendix A shows decomposition results based on Oaxaca-Blinder, which are quantitatively similar to those in the main text.

vant in cases like this, where there are non-linear interactions between different structural forces.

Specifically, I consider the following sets of parameters:

- **Skills:** I fix the skill distribution S_c for the cohorts 2006-10, 2011-15 and 2016-19 at the values estimated for 2001-05.

- **Returns:** I fix the values of $\lambda_{o,\tau}$ and $\eta_{o,\tau}$ for $\tau = 2006..2019$ at the average value estimated for 2001-05. I conduct separate decompositions for each occupation.

- **Demographics:** I fix the distribution of observable characteristics of graduates x_i for the cohorts of 2006-10, 2011-15 and 2016-19 at the level of the 2001-05 cohort.

- **Preferences:** I fix the values of $\omega_{o,\tau}$ for $t = 2006..2019$ at the average value estimated for 2001-05. I fix the values of ρ_c at the value estimated for the 2001-05 cohort.

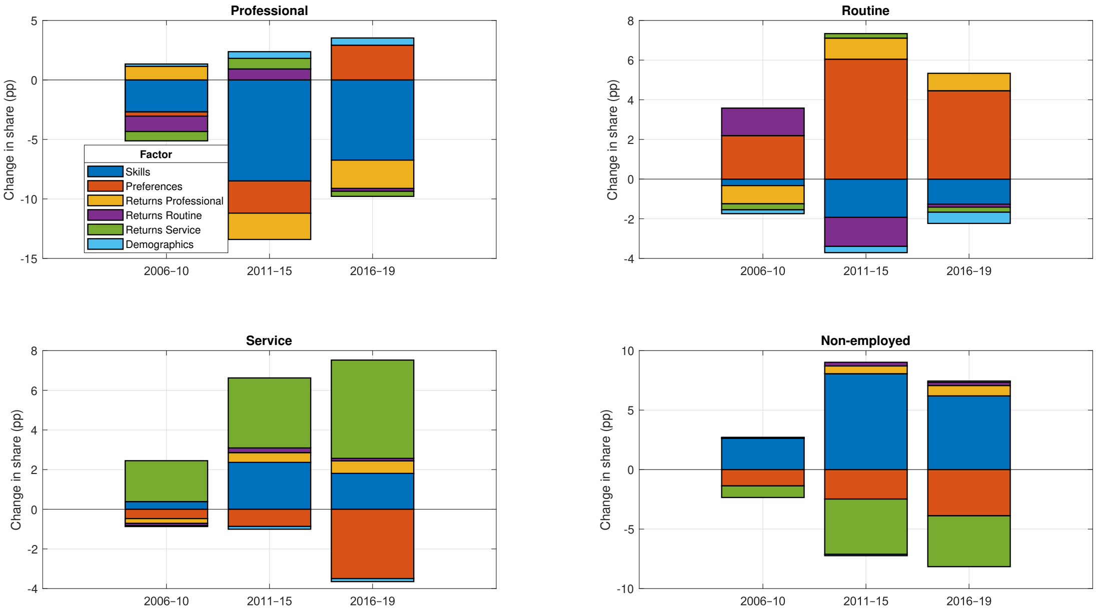
Figure 12 visualises how the different decomposition channels affect the occupation shares of the different cohorts. It can be seen that different factors have different magnitudes, and can affect the observed changes in different directions. For example, changes in the distribution of skill can push the share of professional occupations down, while at the same time returns to skill exert an effect in the opposite direction, leading to an overall more muted change. Factors can also have different impacts for different cohorts highlighting the dynamic nature of these changes. For example, the increase in the returns to skill in routine occupations has a positive effect on the share of graduates entering during the 2006-10 cohort, but a negative one for the 2011-15 cohort.

Inspecting the panels of Figure 12, we note that the most relevant factors appear to be the distribution of skill, the returns to skill in service occupations and the occupation preferences. Returns to skill in professional and routine occupations play an intermediate role particularly for those two occupations, while the impact of changes in the demographic composition appears almost negligible.

In the case of professional occupations, changes in the distribution of skill appear to have the strongest effect in reducing graduate uptake. This is in line with the analysis provided in the previous section: since professional occupations are highly skill intensive, they are not attractive to graduate with a lower level of skill. The downward shifts of the cohort specific skill distributions therefore have a significantly negative effect on the share of professional occupations amongst those cohorts. Table 3 provides additional detail on the Shapley decompositions.[31] The analysis suggests, that by itself, the changing skill distribution between 2001-05 and 2006-10 would have reduced the share of professional

[31] For the decomposition results for log hourly wages see Table A3 in Appendix A.

Figure 12: Shapley Decompositions



Note: Shapley decomposition of changes in occupation shares relative to 2001-05. *Source:* Counterfactual Models.

occupations by 2.7 pp - roughly 70% of the total reduction predicted by the model.[32] In 2011-15 the fall in the share is considerably larger - 8.5 pp, or 76% of the total - while in 2016-19 the difference in the skill distribution is responsible for a 6.7 pp drop, slightly more than the entire reduction relative to 2001-05.

Returns to skill have a still more differentiated impact. Between 2001 and 2010 returns in professional occupations grow rapidly, with the decomposition predicting that by itself this would have increase the share of graduates in professional occupations by around 1.2 pp in 2006-10. However at the same time, as noted previously, returns in routine occupations also grew, offsetting this effect almost exactly - this highlights how the relative balance of returns interacts to create the observed outcomes. For the latter two cohorts, the impact of returns to skill have a decisively negative impact on the share of graduates in professional occupations, on the order of 2.2 and 2.4 pp respectively. Preferences also become quantitatively important for the 2011-15 and 2016-19 cohorts of graduates, with a counterfactual reduction of the graduate share by an additional 2.7 pp during 2011-15, likely as a result of the closing of the amenity gap by routine occupations. This is followed by a counterfactual increase in the share of up to 2.9 pp during 2016-19, as the gap opens up again.

The lions share of the increase in the share of routine occupations is driven by a

[32]Note that the Shapley decomposition calculate the total difference by comparing the baseline model with the *full* counterfactual (i.e. the case where all parameters are fixed at their 2001-05 values. It can therefore come to slight discrepancies compared to the *raw* baseline model differences reported for example in Table A2.

relative increase in the amenity value of associated jobs. For the cohorts after 2001-05, the model suggests increases 2.2, 6.0 and 4.5 pp respectively as a result of rising non-pecuniary utilities. These large counterfactual increases are counteracted somewhat by the leftward shift of the skill distribution, reducing employment in routine occupations by 1.9 and 1.3 pp in 2011-15 and 2016-19 respectively. Returns to skill also have quite sizeable effects, particularly during 2001-05, where the increase in returns add around 1.4 pp to the share, drawing particularly from the professional group. However during 2011-15, the same factor however reduces the share of routine occupations by an almost equivalent amount.

Service occupations benefit from the combined impact of the shifting skill distribution and falling returns to skill. Since service occupations have the lowest skill intensity, they attract graduates from the lower end of the skill distribution. Hence the increasing share of the left tail during 2011-15 and 2016-19 adds around 2.3 and 1.8 pp to the share of service occupations. Even more significant is the fall in returns to skill within services. After 2005 the gap in returns between professional and routine occupations on the one hand and service occupations on the other hand opens up rapidly, a move that polarizes job opportunities, adding 2.1, 3.5 and 5.0 pp in the cohorts after 2001-05, drawing particularly from the set of graduates that would have chosen to remain outside the labour market otherwise. These inflows are however counteracted by the stagnant level of amenities. In contrast to professional and routine occupations, the non-pecuniary utility for service jobs does not rise but remains relatively constant compared to the outside option. This channel appears particularly strong in the last cohort, where the gap compared to the other occupations is particularly large, with a predicted reduction of around 3.5 pp.

The channels affecting the share of graduates choosing to remain out of the labour force are related to those for service occupations. Initially, a small share of graduates had skills low enough that they would not earn a sufficient amount in any occupation. As this share increased the potential pool of graduates for whom non-employment is the best option has increased, adding a potential 2.6, 8.1 and 6.2 pp in the cohorts following 2001-05. That these shifting skill distributions have not caused graduates to leave the labour force in large numbers is particularly due to the falling returns to skill in service occupations, which provides a feasible low-skill intensity alternative for these graduates. By this process, service occupations absorb most of the pool of potentially non-employed graduates. Finally, the increase in non-pecuniary utility among professional and routine occupations post-2005 provides an additional drain on the population of non-employed graduates.

Overall, the decomposition highlights that no single channel fully explains the observed shifts in graduate occupational outcomes; rather, they arise from the interplay of changes in skills, returns, and preferences. The dominant forces differ by type of occupation, and

Table 3: Shapley Decomposition by Period

Panel A: 2006–10 relative to 2001–05

	Total Δ	Skills		Returns						Demographics		Preferences	
				Professional		Routine		Service					
Metric	Δ	Δ	%	Δ	%	Δ	%	Δ	%	Δ	%	Δ	%
Professional	−3.822	−2.672	69.906	1.150	−30.100	−1.334	34.904	−0.791	20.708	0.194	−5.086	−0.370	9.669
Routine	1.858	−0.335	−18.005	−0.935	−50.307	1.400	75.364	−0.285	−15.350	−0.195	−10.499	2.207	118.797
Service	1.550	0.377	24.348	−0.240	−15.496	−0.103	−6.670	2.050	132.275	−0.048	−3.079	−0.486	−31.378
Non-employed	0.415	2.629	634.169	0.024	5.874	0.037	8.964	−0.973	−234.685	0.048	11.672	−1.352	−325.993

Panel B: 2011–15 relative to 2001–05

	Total Δ	Skills		Returns						Demographics		Preferences	
				Professional		Routine		Service					
Metric	Δ	Δ	%	Δ	%	Δ	%	Δ	%	Δ	%	Δ	%
Professional	−11.104	−8.495	76.508	−2.200	19.815	0.886	−7.979	0.856	−7.706	0.524	−4.723	−2.674	24.085
Routine	3.693	−1.917	−51.917	1.043	28.256	−1.440	−39.009	0.256	6.943	−0.288	−7.798	6.038	163.525
Service	5.532	2.337	42.240	0.497	8.985	0.225	4.066	3.513	63.500	−0.138	−2.489	−0.902	−16.301
Non-employed	1.880	8.076	429.659	0.660	35.105	0.330	17.536	−4.625	−246.050	−0.099	−5.256	−2.462	−130.993

Panel C: 2016–19 relative to 2001–05

	Total Δ	Skills		Returns						Demographics		Preferences	
				Professional		Routine		Service					
Metric	Δ	Δ	%	Δ	%	Δ	%	Δ	%	Δ	%	Δ	%
Professional	−6.390	−6.742	105.512	−2.428	38.000	−0.258	4.042	−0.486	7.608	0.608	−9.522	2.916	−45.639
Routine	3.070	−1.267	−41.271	0.863	28.108	−0.193	−6.292	−0.196	−6.397	−0.586	−19.078	4.450	144.930
Service	3.955	1.803	45.586	0.669	16.907	0.158	4.000	4.949	125.144	−0.133	−3.355	−3.491	−88.282
Non-employed	−0.635	6.207	−977.298	0.897	−141.174	0.293	−46.173	−4.267	671.819	0.110	−17.319	−3.875	610.145

Note: Simulations based on a representative sample of graduates. *Source:* Baseline Model and Counterfactual Model Simulations.

can change across cohorts. Together, these results underline the importance of jointly considering skill supply, market returns, and amenity values when analysing occupational sorting over time.

7 Conclusion

The formation of human capital and the acquisition of skills lie at the heart of a university education. With an increasing number of graduates in the UK failing to obtain graduate jobs, there is growing public concern that universities are not equipping graduates with the skills they need to succeed in the labour market. In this paper, I have developed an economic model that accounts for the changing skill distribution of graduates, as well as the evolving returns to these skills and the evolving occupation-specific amenities that jointly determine the distribution of graduates across occupations.

The estimated model makes a number of important findings, that help us understand the dynamics of the labour market for young graduates in the UK. First, the estimates suggest that the skill distribution has experienced a steady downward shift since 2001-05, with average skill levels falling by around 42% of a standard deviation. The reduction in average skill, is however mainly driven by a highly skewed left tail, meaning that while there is a notable up-tick in the share of very low skilled graduates, the majority of graduates still has comparable levels of skill to previous cohorts. Second, the return to skill in professional occupations has increased since 2001, leading to more sorting, resulting in more competitive pressure for the perceived top jobs, which can leave medium skilled grads behind. Third, routine occupations have become more closely aligned to professional jobs, both in terms of returns to skill and levels of non-pecuniary amenities, meaning they become more attractive options for graduates who are squeezed out of professional occupations. Fourth, returns to skill in service occupation has collapsed since 2005, creating a route into employment for low skilled graduates.

This paper has sought to address a small part of a larger research question, and as such necessarily leaves many questions unanswered, some of which suggest themselves as extensions or variants of the model explored here. First, while I provide estimates of the graduate skill distribution, I remain agnostic about the causes of its drift over time. Investigating the drivers of the changing graduate skill distribution, including selection into university education, is likely to be a fruitful avenue for future research. Factors such as changing access to higher education, demographic shifts, and economic incentives all potentially contribute to changes in the skill composition of graduates, and understanding these drivers is key to developing effective policy interventions. Second, while the increasing presence of graduates in non-traditional roles is well documented, there is little research on how this presence influences these occupations. A larger potential pool of highly skilled applicants might facilitate technology adoption within these roles, contributing to the increase in skill prices estimated for these occupations in this paper. Moreover, the increased presence of graduates in these roles could lead to a transformation of the nature of the work itself, as employers adjust job responsibilities to better match the capabilities of their more highly skilled workforce. Understanding these dynamics could help policymakers and industry leaders maximise the benefits of a highly educated workforce, even in sectors not traditionally associated with graduate employment. I leave these and further questions for future research.

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A Additional Tables & Figures

A.1 Additional Tables

Table A1: Overall Model Fit

Share (%)	**Data**	**Model**	**Diff.**	**Diff.(%)**
Professional	61.919	61.599	-0.320	-0.517
Routine	16.607	16.918	0.311	1.871
Service	9.476	9.834	0.357	3.77
Non-employed	11.997	11.65	-0.348	-2.899
Log Wage	**Data**	**Model**	**Diff.**	**Diff.(%)**
Mean Overall	2.536	2.51	-0.026	-1.017
Mean Professional	2.667	2.623	-0.044	-1.632
Mean Routine	2.276	2.314	0.038	1.68
Mean Service	2.13	2.138	0.008	0.365
Variance Overall	0.152	0.15	-0.001	-0.914
Variance Professional	0.113	0.127	0.014	12.675
Variance Routine	0.107	0.109	0.002	2.094
Variance Service	0.09	0.086	-0.004	-4.041
P10 Overall	2.027	2.021	-0.006	-0.308
P10 Professional	2.255	2.189	-0.066	-2.927
P10 Routine	1.902	1.916	0.014	0.717
P10 Service	1.821	1.77	-0.051	-2.819
P50 Overall	2.557	2.525	-0.032	-1.233
P50 Professional	2.671	2.634	-0.037	-1.397
P50 Routine	2.264	2.328	0.064	2.847
P50 Service	2.11	2.147	0.036	1.721
P90 Overall	3.021	2.984	-0.037	-1.241
P90 Professional	3.086	3.056	-0.031	-0.994
P90 Routine	2.681	2.703	0.022	0.818
P90 Service	2.501	2.499	-0.002	-0.091
P50-P10 Overall	0.53	0.505	-0.025	-4.771
P50-P10 Professional	0.416	0.445	0.029	6.901
P50-P10 Routine	0.362	0.412	0.051	14.052
P50-P10 Service	0.289	0.377	0.088	30.312
P90-P50 Overall	0.464	0.458	-0.006	-1.285
P90-P50 Professional	0.415	0.422	0.007	1.599
P90-P50 Routine	0.417	0.375	-0.043	-10.189
P90-P50 Service	0.391	0.352	-0.039	-9.879

Note: Model simulations based on a sample of graduates.
Source: QLFS (2001–2019) and Baseline Model.

Table A2: Differences relative to 2001–05

	2006–10 vs 2001–05				**2011–15 vs 2001–05**				**2016–19 vs 2001–05**			
Share (%)	**Δ (Data)**	**Δ% (Data)**	**Δ (Model)**	**Δ% (Model)**	**Δ (Data)**	**Δ% (Data)**	**Δ (Model)**	**Δ% (Model)**	**Δ (Data)**	**Δ% (Data)**	**Δ (Model)**	**Δ% (Model)**
Professional	-3.667	-5.534	-3.469	-5.212	-9.329	-14.079	-11.344	-17.044	-4.716	-7.117	-5.196	-7.807
Routine	1.351	9.186	1.686	11.284	3.832	26.061	3.808	25.488	2.978	20.25	2.764	18.501
Service	1.891	25.579	1.178	15.817	3.999	54.097	5.578	74.91	2.72	36.798	3.142	42.198
Non-employed	0.425	3.648	0.605	5.471	1.497	12.861	1.957	17.706	-0.983	-8.439	-0.711	-6.43
Log Wage	**Δ (Data)**	**Δ% (Data)**	**Δ (Model)**	**Δ% (Model)**	**Δ (Data)**	**Δ% (Data)**	**Δ (Model)**	**Δ% (Model)**	**Δ (Data)**	**Δ% (Data)**	**Δ (Model)**	**Δ% (Model)**
Mean	-0.025	-0.957	-0.047	-1.835	-0.113	-4.373	-0.148	-5.735	-0.072	-2.801	-0.11	-4.25
Variance	0.004	2.615	0.008	5.76	0.004	2.947	0.007	4.659	-0.01	-6.366	0.007	4.796
P10	-0.043	-2.07	-0.061	-2.922	-0.123	-5.938	-0.163	-7.746	-0.042	-2.038	-0.113	-5.385
P50	-0.022	-0.856	-0.04	-1.525	-0.111	-4.255	-0.137	-5.294	-0.082	-3.131	-0.098	-3.764
P90	-0.01	-0.319	-0.049	-1.586	-0.091	-2.955	-0.155	-5.046	-0.091	-2.982	-0.136	-4.423
Mean Professional	-0.01	-0.352	-0.035	-1.293	-0.08	-2.941	-0.108	-4.037	-0.082	-3.018	-0.101	-3.764
Mean Routine	0.017	0.741	-0.015	-0.624	-0.046	-2.018	-0.076	-3.241	0.058	2.533	-0.05	-2.126
Mean Service	-0.015	-0.706	-0.065	-2.958	-0.029	-1.369	-0.098	-4.492	0.025	1.148	-0.035	-1.59

Note: Simulations based on a representative sample of graduates. *Source:* QLFS (2001-19) and Baseline Model.

Table A3: Shapley Decomposition by Period - Log Hourly Wages

Panel A: 2006–10 relative to 2001–05

	Total Δ	**Skills**		**Returns**						**Demographics**		**Preferences**	
				Professional		**Routine**		**Service**					
Metric	Δ	Δ	%	Δ	%	Δ	%	Δ	%	Δ	%	Δ	%
Mean Overall	−0.032	−0.052	163.161	0.027	−83.292	0.004	−11.486	−0.002	6.157	−0.001	2.975	−0.007	22.486
Mean Professional	−0.019	−0.052	275.886	0.035	−189.109	−0.005	27.984	0.007	−35.761	−0.002	10.039	−0.002	10.961
Mean Routine	0.005	−0.042	−783.092	−0.018	−339.000	0.068	1268.922	0.008	148.997	−0.003	−57.475	−0.007	−138.352
Mean Service	−0.039	−0.031	80.048	−0.009	22.274	−0.003	7.225	0.016	−39.643	0.000	−0.104	−0.012	30.200

Panel B: 2011–15 relative to 2001–05

	Total Δ	**Skills**		**Returns**						**Demographics**		**Preferences**	
				Professional		**Routine**		**Service**					
Metric	Δ	Δ	%	Δ	%	Δ	%	Δ	%	Δ	%	Δ	%
Mean Overall	−0.131	−0.081	62.183	−0.019	14.320	−0.001	0.616	−0.012	9.044	0.005	−3.718	−0.023	17.555
Mean Professional	−0.089	−0.071	79.965	−0.012	13.918	−0.008	8.978	0.009	−9.653	0.003	−3.883	−0.010	10.675
Mean Routine	−0.060	−0.059	98.219	−0.013	20.917	0.008	−13.113	0.012	−20.562	0.001	−2.082	−0.010	16.622
Mean Service	−0.092	−0.023	25.154	−0.002	2.448	−0.003	2.775	−0.058	63.518	0.002	−1.676	−0.007	7.782

Panel C: 2016–19 relative to 2001–05

	Total Δ	**Skills**		**Returns**						**Demographics**		**Preferences**	
				Professional		**Routine**		**Service**					
Metric	Δ	Δ	%	Δ	%	Δ	%	Δ	%	Δ	%	Δ	%
Mean Overall	−0.104	−0.074	71.084	−0.006	5.933	0.003	−2.727	−0.005	4.946	0.002	−1.943	−0.024	22.706
Mean Professional	−0.088	−0.069	78.946	0.011	−12.374	−0.011	12.033	0.012	−14.093	0.001	−0.778	−0.032	36.266
Mean Routine	−0.042	−0.050	120.818	−0.034	80.618	0.060	−143.474	0.018	−42.498	−0.004	10.717	−0.031	73.818
Mean Service	−0.028	−0.030	106.052	−0.006	20.549	−0.004	12.455	0.042	−148.861	−0.003	9.373	−0.029	100.431

Note: Simulations based on a representative sample of graduates. *Source:* Baseline Model and Counterfactual Model Simulations.

Table A4: μ: Skill-bin mass by cohort

	2001–05	2006–10	2011–15	2016–19
Bin 1	-0.013	1.168	3.294	2.677
	(0.001)	(0.002)	(0.003)	(0.003)
Bin 2	-3.694	-2.884	-2.121	-1.666
	(0.006)	(0.009)	(0.018)	(0.019)
Bin 3	-5.454	-4.124	-4.414	-2.602
	(0.009)	(0.022)	(0.030)	(0.015)
Bin 4	-4.464	-1.110	-0.730	-3.244
	(0.005)	(0.005)	(0.004)	(0.014)
Bin 5	-1.864	-0.003	0.283	0.230
	(0.001)	(0.002)	(0.003)	(0.004)
Bin 6	-0.217	0.822	2.006	1.341
	(0.001)	(0.002)	(0.003)	(0.004)
Bin 7	1.378	2.281	3.340	2.802
	(0.001)	(0.002)	(0.003)	(0.004)
Bin 8	2.377	3.257	4.399	4.195
	(0.001)	(0.002)	(0.003)	(0.004)
Bin 9	1.485	2.040	3.050	2.518
	(0.001)	(0.002)	(0.003)	(0.004)
Bin 10	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)

Note: Parameter estimates. Numerical standard errors in parentheses.
Source: Baseline Model.

Table A5: Fixed effects (η), Returns to skill (λ), and Preferences (ω) by occupation

	Fixed effects (η)			**Returns** (λ)			**Preferences** (ω)		
Year	Professional	Routine	Service	Professional	Routine	Service	Professional	Routine	Service
2001	-0.123	0.195	0.350	3.199	2.397	2.142	-2.164	-1.888	-2.031
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
2002	-0.194	0.065	0.004	3.329	2.607	2.569	-2.161	-1.880	-2.003
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
2003	-0.210	-0.102	0.164	3.365	2.838	2.431	-2.193	-1.943	-2.041
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
2004	-0.324	-0.219	0.195	3.535	3.026	2.381	-2.171	-1.922	-1.999
	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
2005	-0.098	-0.097	0.575	3.269	2.817	1.827	-2.230	-1.909	-1.999
	(0.000)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
2006	-0.334	-0.285	-0.024	3.563	3.121	2.691	-2.174	-1.853	-1.900
	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
2007	-0.110	-0.215	0.638	3.276	3.019	1.822	-2.131	-1.742	-1.903
	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)	(0.001)
2008	-0.410	-0.379	1.467	3.683	3.267	0.702	-2.097	-1.750	-2.008
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
2009	-0.548	-0.291	1.269	3.845	3.122	0.954	-2.159	-1.855	-2.023
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
2010	-0.697	-0.613	1.268	4.036	3.567	0.957	-2.108	-1.782	-1.982
	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)	(0.001)
2011	-0.632	-0.529	1.740	3.919	3.375	0.101	-2.219	-1.844	-2.043
	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)
2012	-0.411	-0.457	1.788	3.610	3.248	0.001	-2.076	-1.674	-1.963
	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)
2013	-0.742	-0.346	1.794	4.025	3.111	0.057	-1.997	-1.594	-1.949
	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)
2014	-0.375	-0.740	1.766	3.510	3.656	0.001	-1.928	-1.591	-1.893
	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)
2015	-0.659	-0.486	1.762	3.913	3.303	0.000	-1.924	-1.541	-1.944
	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)
2016	-0.574	-0.453	1.807	3.817	3.287	0.000	-1.922	-1.562	-1.942
	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)
2017	-0.722	-0.525	1.787	4.005	3.412	0.080	-1.748	-1.544	-1.995
	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
2018	-0.743	-0.754	1.794	4.033	3.698	0.280	-1.803	-1.511	-2.024
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
2019	-1.882	-1.062	1.749	5.569	4.137	0.385	-1.694	-1.433	-1.957
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)

Note: Parameter estimates. Numerical standard errors in parentheses.
Source: Baseline Model.

Table A6: Additional parameters

Panel A: ρ (cohort scaler)

	2001–05	2006–10	2011–15	2016–19
ρ	3.236	3.409	3.011	2.807
	(0.002)	(0.002)	(0.002)	(0.001)

Panel B: ϕ (idiosyncratic shocks by occupation and year)

Occupation	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Professional	0.126	0.062	0.085	0.082	0.093	0.072	0.117	0.076	0.093	0.081	0.064	0.141	0.073	0.113	0.070	0.073	0.079	0.117	0.067
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Routine	0.147	0.105	0.092	0.071	0.109	0.072	0.078	0.050	0.087	0.060	0.062	0.064	0.101	0.073	0.081	0.049	0.064	0.067	0.094
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Service	0.200	0.087	0.104	0.090	0.152	0.075	0.158	0.184	0.144	0.172	0.277	0.248	0.268	0.245	0.274	0.245	0.222	0.274	0.323
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

Panel C: Observables on wages (β)

Occupation	Sex	Dgr	Exp	Exp^2
Professional	-0.074	0.013	0.072	-0.002
	(0.000)	(0.000)	(0.000)	(0.000)
Routine	-0.036	0.004	0.098	-0.004
	(0.000)	(0.000)	(0.000)	(0.000)
Service	-0.023	0.016	0.096	-0.005
	(0.000)	(0.000)	(0.000)	(0.000)

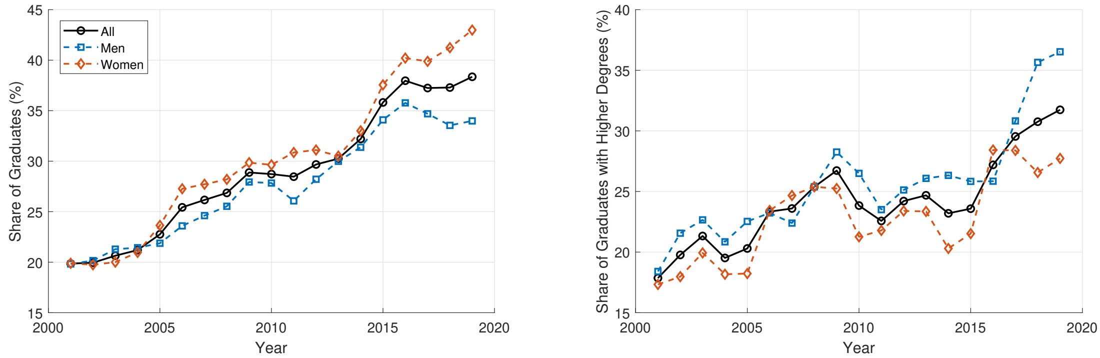
Panel D: Observables on non-pecuniary utility (γ)

Occupation	Sex	Dgr	Exp	Exp^2
Professional	-0.007	-0.026	0.165	-0.012
	(0.000)	(0.000)	(0.001)	(0.000)
Routine	-0.029	-0.226	0.018	-0.003
	(0.000)	(0.000)	(0.001)	(0.000)
Service	0.139	-0.223	-0.033	-0.000
	(0.000)	(0.000)	(0.001)	(0.000)

Note: Parameter estimates. Numerical standard errors in parentheses.
Source: Baseline Model.

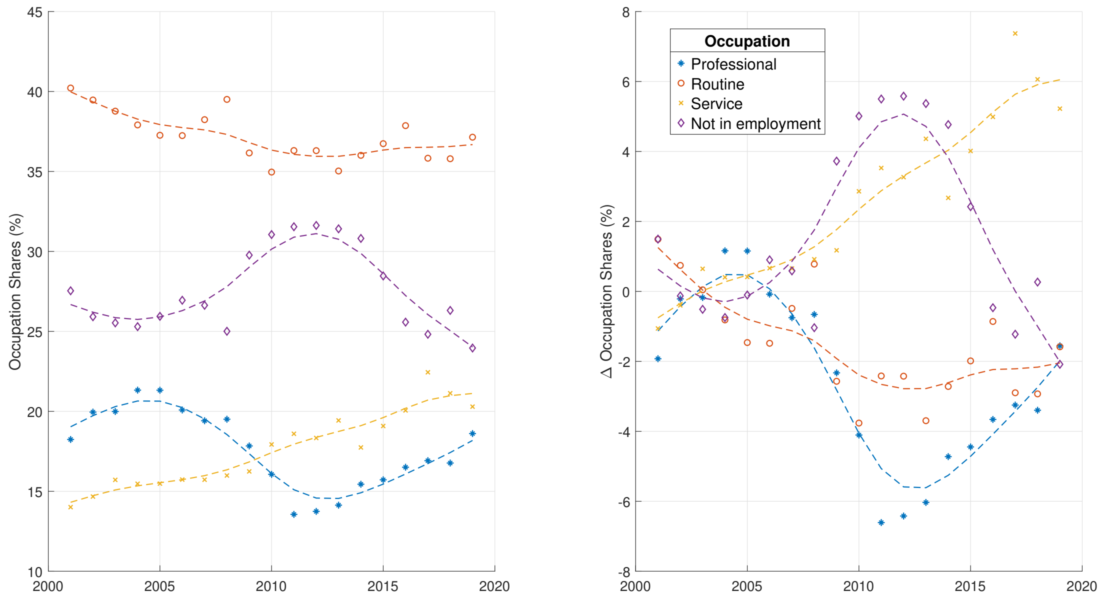
A.2 Additional Figures

Figure A1: Trends in higher education attainment among young adults in the UK



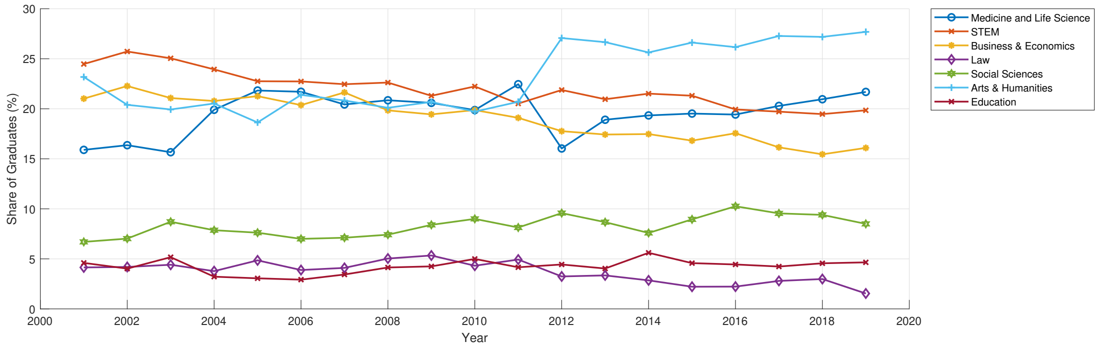
Note: Subplot 1: Share of young adults with a university degree, aged 21-30 years. Subplot 2: Share of graduates with a higher degree, aged 21-30 years. *Source:* Quarterly Labour Force Survey (2001-2019).

Figure A2: Trends in occupation shares of young adults without a degree in the UK



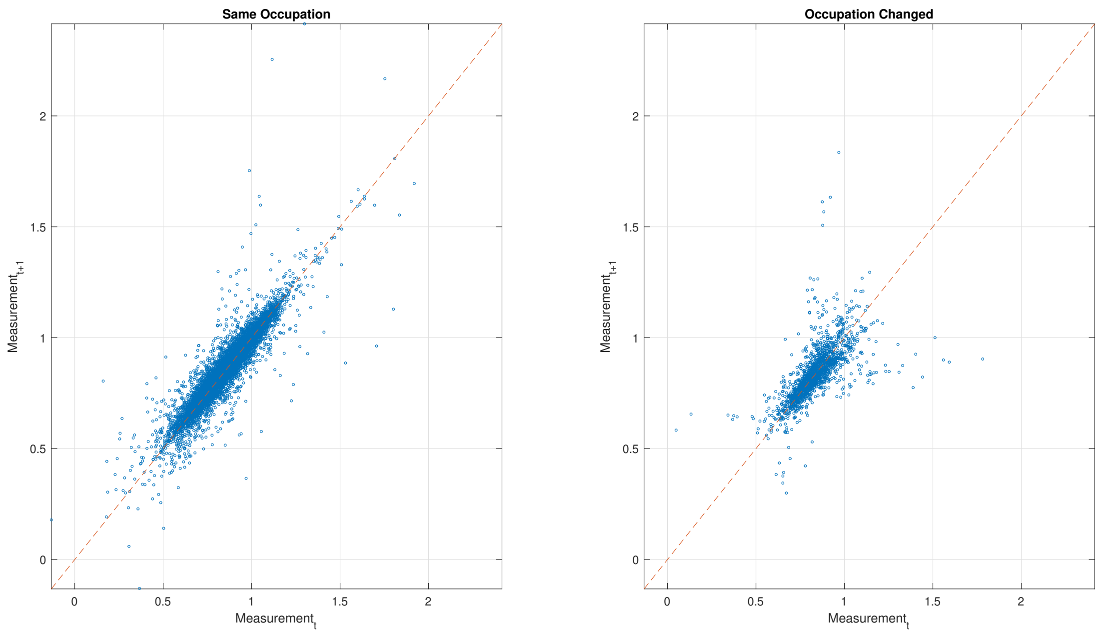
Note: Adults without a university degree, aged 21-30 years; change relative to 2001-05 average. Broken lines are HP-filtered trends (smoothing parameter = 6.25). SOC 2000 1-Digit Occupation Classification. Professional includes codes 1-3; Routine includes codes 4,5,8,9; Service includes codes 6,7; Not in employment includes unemployed and those out of the labour force for other reasons. *Source:* Quarterly Labour Force Survey (2001-2019).

Figure A3: Trends in primary subject studied by university graduates in the UK



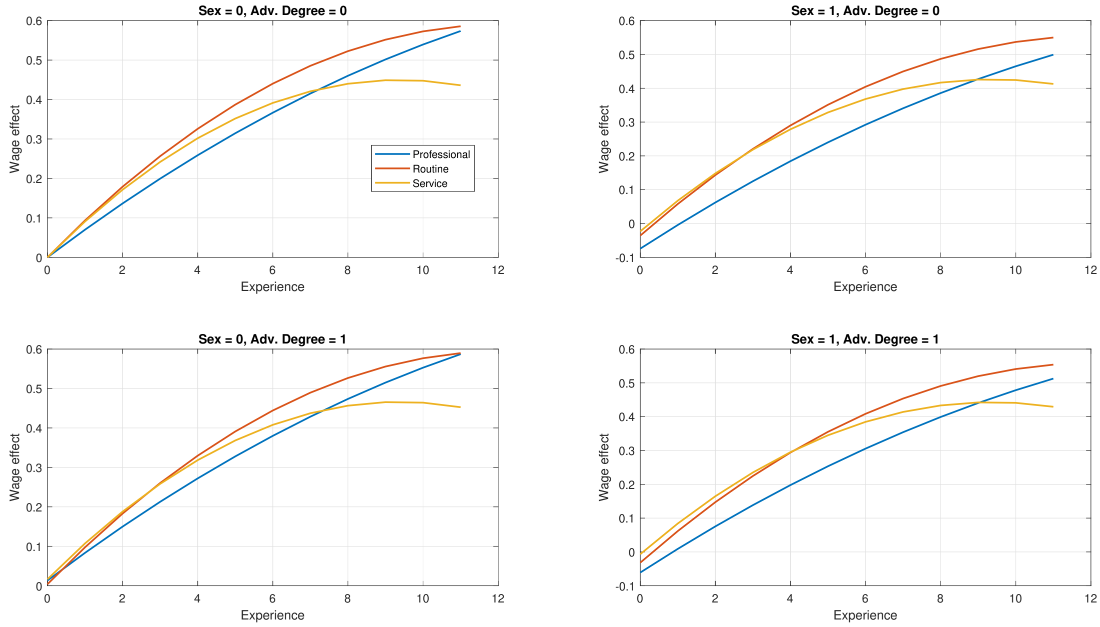
Note: Adults without a university degree, aged 21-30 years; Single subject degrees only. *Source:* Quarterly Labour Force Survey (Cross-sectional) (2001-2019).

Figure A4: Correlation of consecutive skill measurements



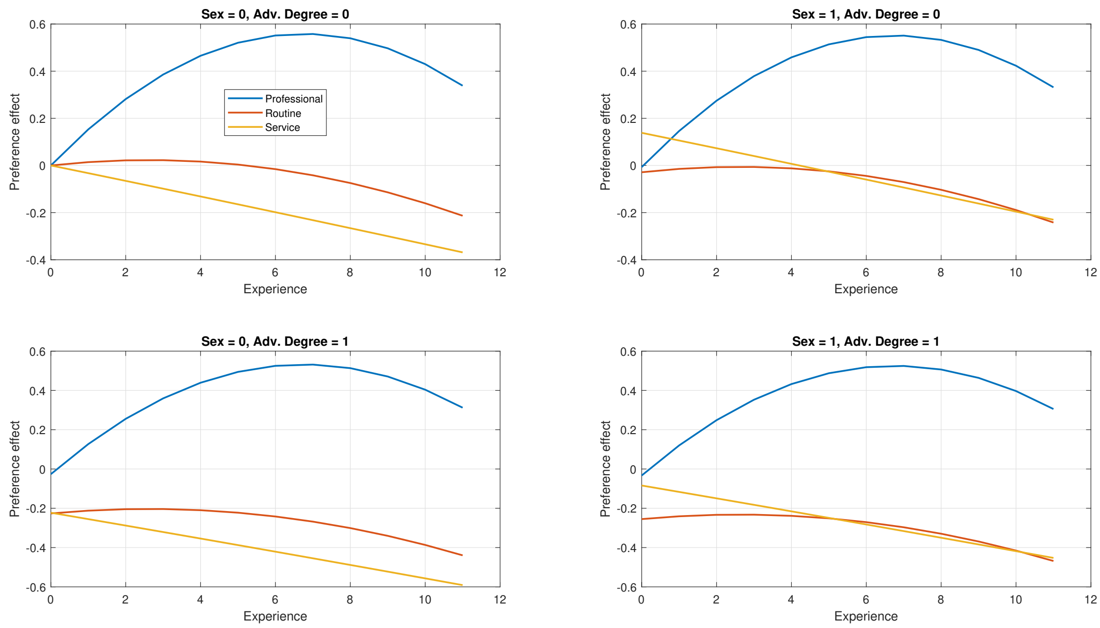
Note: The figure plots the consecutive skill measures for each individual in the data. Skill measures are obtained by calculating $s_{i,t} = \frac{w_{i,t} - \eta_{i,t}}{\lambda_{i,t}}$, based on the estimated values of η and λ. To avoid exploding measurements, I drop observations with $\lambda < 0.5$. For more details, see Diegert, 2024.
Source: QLSF Data and Baseline Model.

Figure A5: Lifecycle Effects: Wages



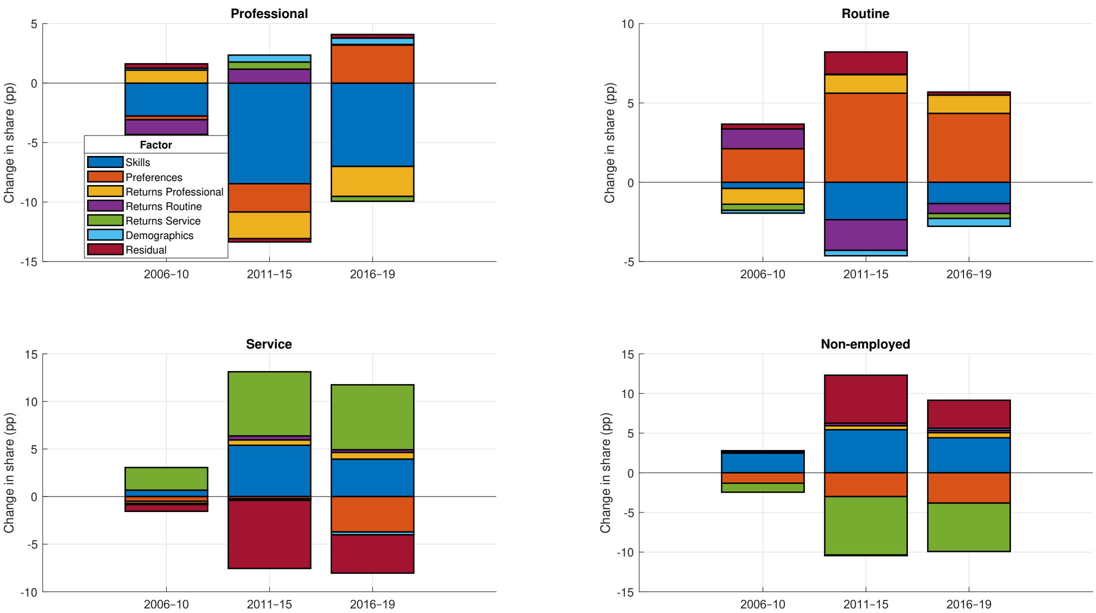
Note: Contribution of demographic characteristics to log wage. Sex = 0 denotes male graduates, Sex = 1 denotes female graduates. *Source:* Baseline Model.

Figure A6: Lifecycle Effects: Non-wage Amenities



Note: Contribution of demographic characteristics to non-pecuniary part of utility. Sex = 0 denotes male graduates, Sex = 1 denotes female graduates. *Source:* Baseline Model.

Figure A7: Oaxaca-Blinder Decompositions



Note: Oaxaca-Blinder decomposition of changes in occupation shares relative to 2001-05. *Source:* Counterfactual Models.

B Technical Details

B.1 Imputation of missing wage values

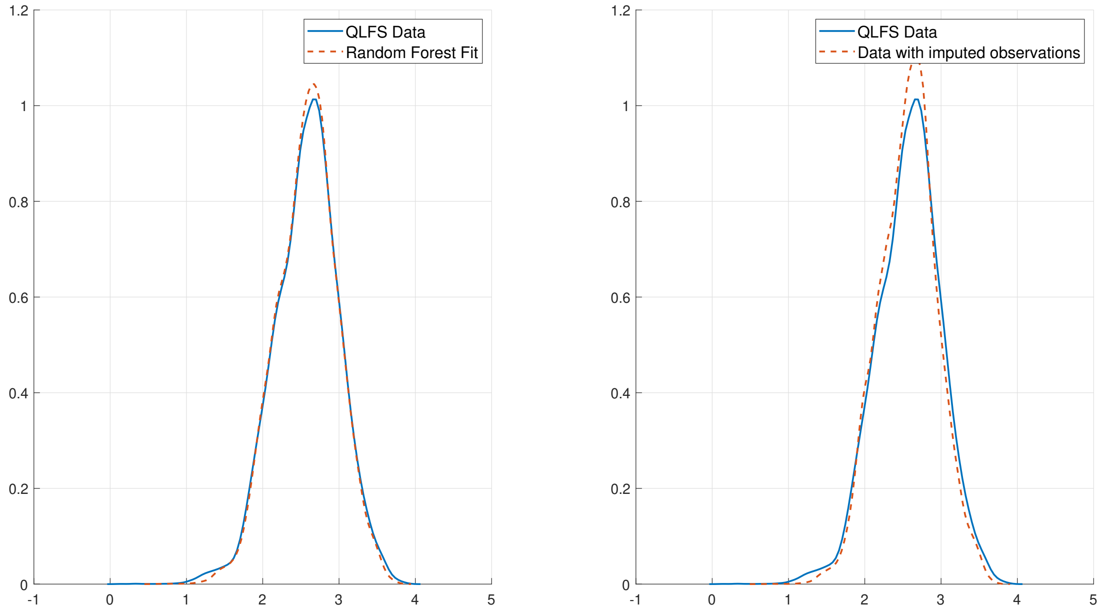
In the QLFS, wage data is only collected during the first and last interviews, resulting in three missing wage observations for each individual. To impute these missing wage values, I employ a two-step methodology that leverages both fixed effects to account for individual-specific skills and a Random Forest model to predict the missing wages.

A key problem for the imputation of wage data based on observable characteristics is that these methods do not account for unobservable heterogeneity among individuals, which is key in a panel setting. To account for individual-specific abilities, I incorporate individual fixed effects into the imputation procedure. This is done by specifying a linear fixed-effects regression where the hourly pay is regressed on variables such as experience, its square (to capture non-linear effects), year, quarter, sex, occupation, and government office region, including a fixed effect for each individual. This model helps us extract individual-specific effects, which are then used in the second stage of the imputation procedure. For the imputation phase, I use a Random Forest regression model. The predictors include the fixed effects extracted from the fixed-effects model along with experience, year, quarter, sex, occupation, and government office region.

The Random Forest model is configured with 200 trees, allowing for robust predictions by averaging over multiple decision trees to reduce overfitting. I ensure that the model is fine-tuned by setting the maximum number of splits to the number of observations in the training data minus one and a minimum leaf size of one. This configuration helps in capturing the complex relationships within the data. The predicted wages are then integrated back into the dataset, replacing the missing values.

Figure B1 below showcases the results of the imputation procedure. The first panel depicts the observed wages, as well as the predictions based on the random forest model. The close alignment between the actual data and the predicted values indicates a good fit of the model to the observed values. This suggests that the model captures the underlying patterns in the wage data effectively, validating the robustness of the imputation methodology. The second panel shows the distribution of hourly wages with and without imputed values. The overall shape of the distribution remains consistent, demonstrating that the imputed values align well with the observed data distribution. This suggests, that the imputed values are not arbitrary but rather grounded in the underlying data patterns, preserving the integrity of the dataset.

Figure B1: Distribution of observed wages and imputed values



Note: Kernel density estimates of CPI (2014) deflated log hourly wages. *Source:* QLFS (2001-2019) and authors' calculations.

B.2 Estimation algorithm

The estimation procedure is a simple application of simulated maximum likelihood. The objective is to find a vector of parameters θ that maximises the probability of observing the actual outcomes. The only complication, that arises in this case is that we do not have a closed-form solution for the joint probability (9) and thus have to approximate the integral via quadrature or simulation.

Take graduate i with skills s_i, then, the conditional joint probability of observing the occupation choice $o^*_{i,t}$ and the wage $w^*_{i,t}$ is given by 7. Since we observe graduates for $T = 5$ periods the probability of observing the sequences o^*_i and w^*_i is given by $\Pr(o^*_i, w^*_i|s_i) = \prod_{t=1}^{T} \Pr(o^*_{i,t}, w^*_{i,t}|s_i)$. Following from K. Train, 2016, the unconditional probability can be approximated by: $\Pr(o^*_i, w^*_i) = \sum_{r=1}^{R} \Pr(o^*_i, w^*_i|s_r)W(s_r|\alpha)$, where $W(s_r|\alpha)$ is a weighting scheme that describes the probability of drawing s_r from some distribution S and α is a vector of parameters that describes that distribution. As described in section 4, I approximate S with a histogram on the unit interval with L evenly spaced bins B_l. Hence, for a random s_r, the probability $W(s_r|\alpha)$ is equal to the probability of selecting bin B_l, into which s_r falls. I specify $Pr(B_l) = \frac{e^{\alpha_l}}{\sum_{l=1}^{L} e^{\alpha_l}}$, with α_L normalised to 0. This procedure provides a fast, and importantly numerically stable way of calculating the weights and hence approximate $\Pr(o^*_i, w^*_i)$. Standard results suggest, that as long as one uses a large enough number of draws to approximate the integral, the Maximum Simulated Likelihood Estimation (MSLE) is asymptotically equivalent to classical Maximum

Likelihood Estimation (MLE) (c.f. McFadden & K. Train, 2000).

For ease of notation let's assume that α is a subset of the extended parameter vector θ. We can write down the simulated log-likelihood function as:

$$ll^{sim}(\theta) = \frac{1}{N}\sum_{i=1}^{N} ln[\sum_{r=1}^{R}\sum_{o=1}^{O} \mathbf{1}_{(o=o^*)} \Pr(o_i^*, w_i^*|s_r)W(s_r|\alpha)] \tag{B1}$$

and we can estimate θ as:

$$\hat{\theta} = \arg\max_{\theta} ll^{sim}(\theta) \tag{B2}$$

The complete algorithm proceeds as follows:

1. Set $p = 1$ and make a guess for $\hat{\theta}_1$. Specify a tolerance criterion ϵ. Set R, the number of draws used to approximate the integral to a reasonably high number.

2. For each individual i, given $\hat{\theta}_p$ draw a vector of s_i, R times, denoting each as s_i^r.

3. For $r = 1$ to R :

   (a) For $q = 1$ to 5 : Calculate $\nu_{iq}^r = w_{iq}^* - (\eta_{o^*} + \lambda_{o^*}^\tau s_i^r + \beta x_i)$. For a given pair s_i^r, ν_{iq}^r calculate $\Pr_q^r(o_{iq}^*, w_{iq}^*|s_i^r)$.

   (b) Calculate $W(s_r|\alpha)$.

   (c) Calculate $\Pr^r(o_i^*, w_i^*|s_i^r)$ as $\prod_{q=1}^{5} \Pr_q^r(o_{iq}^*, w_{iq}^*|s_i^r)$

4. Evaluate the integral over all R values of $\Pr^r(o_i^*, w_i^*|s_i^r)$ to obtain:

   $\Pr(o_i^*, w_i^*) = \sum_{r=1}^{R} \Pr^r(o_i^*, w_i^*|s_i^r)W(s_r|\alpha)$.

5. Repeat steps $2-4$ for all N individuals. Calculate the log-likelihood via (B1) denoting it as ll_p^{sim}.

6. If $|ll_p^{sim} - ll_{p-1}^{sim}| < \epsilon$, terminate here. Otherwise, increment p and find a new value $\hat{\theta}_p$ and repeat from step 2.

For the numerical evaluation of the integral, I use a grid of 100 quasi-random Halton draws, which have been shown to provide about an order of magnitude more accuracy than simple random draws (K. Train, 2000, K. E. Train, 2009). To ensure stochastic equicontinuity, I use the same set of points for each agent at each iteration. For updating $\hat{\theta}_p$ in step 6, I use Matlab's `fminunc` routine, using central numerical derivatives and critical values of $1e^{-12}$ for convergence.

I calculate numerical standard errors following the well-known (c.f. K. E. Train, 2009) relationship between the Hessian of the likelihood function and the information identity:

For the correctly specified model, the error of the MLE estimate $\hat{\theta}$ is distributed according to:

$$\sqrt{N}(\hat{\theta} - \theta^*) \to N(0, -\mathbf{H}^{-1}) \tag{B3}$$

where θ^* is the true parameter vector, and $-\mathbf{H}$ is the information matrix. To avoid complications due to the numerical procedure and the high dimensionality of the problem, I calculate a numerical Hessian of the likelihood function at the SMLE-estimate and then use a pseudo-inverse (c.f. Gill & King, 2004) to obtain the standard errors for the estimated parameters.

Table B1: Summary of Model Parameters

Parameter	**Description**	**Number of Parameters**
α_c	Parameters describing the skill distribution.	36
$\eta_{o,\tau}$	Occupation-year specific fixed effect.	57
$\lambda_{o,\tau}$	Occupation-year specific return to skill.	57
$\omega_{o,\tau}$	Occupation-year specific occupation preferences.	57
β	Contribution of observables to wage.	12
γ	Contribution of observables to non-wage amenities.	12
ρ_c	Scaler of idiosyncratic preference shock.	4
$\phi_{o,\tau}$	Standard deviation of log wage measurement errors.	57

B.2.1 The cluster refinement global optimisation algorithm

The likelihood function generated by this problem is smooth, but not globally concave, which makes it difficult for gradient-based optimisation routines that are prone to converge to local minima. This is a general problem for the class of discrete choice models, but especially here given the high dimensionality of the parameter space. To maximise the log-likelihood function, I therefore develop a novel global optimisation algorithm that utilises machine learning to effectively search through the high-dimensional parameter space. The algorithm proceeds as follows:

1. Define a grid of initial starting points G that spans the parameter space θ. For each point in G evaluate the log-likelihood function.

2. Discard points where the log-likelihood is below a certain threshold criterion.

3. Use a clustering algorithm to cluster the remaining points into K clusters.

4. From each cluster select a set of points P^k. The selection can either be the point with the best log-likelihood value in the cluster or a weighted average of all points in the cluster or both.

5. Use a local solver starting at each point in P^k to maximise the log-likelihood function using a nonlinear optimisation routine. Repeat steps 2-5 as required.

The main idea behind the algorithm is that the clustering algorithm will group points that are *similar* together. Points that are close together in the parameter space are likely in the neighbourhood of the same local maximum, so it is unnecessary to run local solvers from each of these points. The computational savings can be used to explore further regions of the parameter space.

For further refinement, steps 2-5 can be repeated using the local maxima found by the nonlinear solvers in step 5 and so on. Using this method it is practical to start with a large number of clusters in the beginning and reduce this number in each successive iteration. In doing so, it is advised to initially set the convergence criteria to relatively high values or limit the number of iterations for the local solvers in the beginning and tighten the criteria over successive iterations.

For the estimation, I begin with a grid of 1,000,000 points where the likelihood function is evaluated once on each point. I then run 3 iterations of the cluster refinement algorithm on the set of these points, using a cluster number of 100 in the first round, 20 in the second round and 1 in the third round. In each round, I select two points from each cluster, namely the point with the best likelihood value and the cluster's weighted midpoint. For the final run, I select the point with the best likelihood value.

B.3 Alternative interpretation of ϕ

In section4, I introduced ϕ as the standard deviation of an idiosyncratic shock to the graduates' wage that was assumed to be independent of the graduates' occupation choice. In this subsection, I want to quickly outline an alternative interpretation of ϕ that doesn't rely on the structural interpretation and therefore might be easier to accept for some readers.

To illustrate, let us return to the joint probability (9):

$$\Pr(o_i^*, w_i^*) = \int \Pr(o_i^*|s_i) \Pr(w_i^*|s_i, o_i^*) f(s) d(s).$$

Note that this formulation shows that the estimation is essentially trying to match two conditional probabilities: i) the conditional probability of choosing occupation o_i^*; ii) the conditional probability of the observed distribution of wages w_i^*. The hope is that if the model is flexible enough (i.e. has enough free parameters), there will be no conflict between these two objectives: the same parameter vector θ^* that maximises the joint probability also maximises the individual conditional probabilities. However, in reality, we might not be close to θ^* and particularly during the estimation, the estimator will encounter points where trade-offs have to be made between the two counteracting objectives. In other words, the estimator needs to have an exchange rate to trade off a better fit on one dimension against a worse fit on another.

By looking at the way that ϕ enters the likelihood function to see that it provides an implicit weight for making this trade-off: $\frac{e^{(-\frac{\nu_i^2}{2\phi^2})}}{\sqrt{2\pi\phi^2}}$. If ϕ is small, then values of ν_i away from 0 will lead to large losses in terms of likelihood. In other words, there is a high priority on matching the wage distribution, even at the expense of the occupation distribution. If ϕ is large, then the estimator is more forgiving with respect to large deviations from the observed wage and puts relatively more weight on matching the conditional occupation choice probabilities. In this interpretation, ϕ is simply a tuning parameter that helps us find the right balance among our different objectives.